



PAN3029 / PAN3060 系列

产品说明书

Sep. 2023
V1.4

Confidential

上海磐启微电子有限公司

低功耗远距离无线收发芯片

summarize

The PAN3029 / PAN3060 is a low-power long-distance wireless transceiver with ChirpIoT™ modulation and demodulation technology. Supporting half-duplex wireless communication and operating in the frequency bands of 408~565MHz and 816~1080MHz, the chip features high immunity to interference, high sensitivity, low power consumption, and ultra-long-distance transmission. With a maximum sensitivity of -143dBm and maximum output power of 20dBm, the chip produces industry-leading link budgets, making it the best choice for long-distance transmission and applications that require high reliability. The PAN3029 / PAN3060 also offers significant advantages over conventional modulation techniques in terms of blocking and neighbor channel selection, which can further improve communication reliability. At the same time, it also offers greater flexibility in terms of user-adjustable spread spectrum modulation bandwidth, spreading factor and error correction rate, effectively improving the trade-off between distance, interference immunity and power consumption for chips using conventional modulation techniques.

Main characteristics

- Operating frequency band: 408~565MHz, 816~1080MHz
- Modulation: ChirpIoT™
- Transmitter output power: -30dBm~20dBm
- Maximum link budget up to: 163dB
- Sensitivity down to: -143dBm@62.5kHz
- Operating Current
 - Sleep current: 200nA
- Receiving current: 4.1mA@DCDC mode
 - Emission current:
 - 91mA@20dBm
 - 13mA@-2dBm
- Supported Bandwidth
 - PAN3029: 62.5kHz, 125kHz, 250kHz, 500kHz
 - PAN3060: 125kHz, 250kHz, 500kHz
- Support SF factor, support spread spectrum factor automatic identification
 - PAN3029: 5 to 12
 - PAN3060: 5~9
- Supported code rates: 4/5, 4/6, 4/7, 4/8
- CAD support
 - Support low rate mode: 0.08~59.9kbps
- Supports 4-wire SPI / 3-wire SPI / I2C configuration interface, supports 4 GPIOs

- Operating Voltage: 1.8V ~ 3.6V (DCDC mode 2V ~ 3.6V)
- Working temperature: -40°C~85°C
- Package: QFN28 (4 x 4mm)

typical application

- Smart Factory
- Smart Water
- intelligent agriculture
- intelligent medical care
- intelligent community
- intelligent firefighting (psychology)

Transceivers catalogs

Overview.....	2
Main Characteristics.....	2
Typical Applications.....	2
Catalog.....	3
1 Naming rules	6
2 Ordering Information.....	7
3 System Architecture Block Diagram.....	8
4 Pin Definitions and Descriptions.....	9
4.1 Pinout.....	9
4.2 Pin Description.....	10
5 Electrical Characteristics.....	12
5.1 Absolute Maximum Ratings.....	12
5.2 DC Characteristics.....	12
5.3 RF Characteristics.....	14
6 Frequency synthesizer circuit.....	17
7 Modem	18
7.1 Modulation Parameters.....	18
7.1.1 Signal Bandwidth.....	18
7.1.2 Spread Spectrum Factor.....	18
7.1.3 Code rate.....	18
7.1.4 Low Rate Mode.....	18
7.2 Frame Structure.....	19
8 MAC Design.....	20
8.1 Transmit/Receive Mode.....	20
8.2 Transmitter Mode.....	20
8.2.1 Single Packet Transmission Mode.....	21
8.2.2 Continuous Transmit Mode.....	22
8.3 Receiver Mode.....	23
8.3.1 Single Packet Receive Mode.....	23
8.3.2 Single packet receive mode with timeout.....	24
8.3.3 Continuous reception mode.....	25
8.4 IRQ Interrupt.....	26
8.5 CAD-IRQ Interrupt.....	27
9 Interface Design.....	28
9.1 Four-wire SPI.....	28
9.1.1 4-Wire SPI Write Timing.....	29
9.1.2 4-Wire SPI Read Timing.....	29
9.1.3 Four-Wire SPI Timing Requirements.....	29
9.2 Three-wire SPI.....	30
9.2.1 Three-wire SPI Write Timing.....	30
9.2.2 3-Wire SPI Read Timing.....	30
9.2.3 Three-Wire SPI Timing Requirements.....	30
9.3 I2C.....	31
9.3.1 I2C Write Timing.....	31
9.3.2 I2C Read Timing.....	31
9.3.3 I2C Timing Requirements.....	32

		Transceivers	
9.4	FIFO		32
9.5	GPIO		32
9.5.1	External PA Control GPIO		32
9.5.2	CAD Monitoring GPIO		33
10	ChirpIoT Special Feature Description		34
10.1	RSSI Functions		34
10.1.1	RSSI indication of ChirpIoT signals		34
10.1.2	Channel Detection RSSI Indication		34
10.2	Signal Noise Energy Statistics Function		34
10.3	Channel Active Detection (CAD)		34
10.3.1	ChirpIoT signaling lead segment detection		35
10.3.2	ChirpIoT signaling data segment detection		36
10.4	MAPM Mode Function		36
10.5	eFuse Features		37
10.6	Early Interruption Function		37
10.7	Intelligent SF Recognition		38
11	Operating Mode Description		39
11.1	Overview		39
11.2	Operating Mode Description		40
11.2.1	Deep Sleep Mode		40
11.2.2	Sleep Mode		40
11.2.3	STB1 Mode		40
11.2.4	DCDC Mode		40
11.2.5	STB2 Mode		40
11.2.6	STB3 Mode		40
11.2.7	TX Mode		40
11.2.8	RX Mode		40
12	Registers		41
12.1	System Configuration		41
12.2	Mode Configuration		41
12.3	MAC Configuration (Low Voltage Area Register)		42
12.4	GPIO Interface Configuration (Low Voltage Register)		43
12.5	Basic Operation Configuration		44
13	Reference Schematic		46
14	Package Size		49
15	Cautions		50
16	Storage conditions		51
	Abbreviations		52
	Revision History		53
	联系方式		54

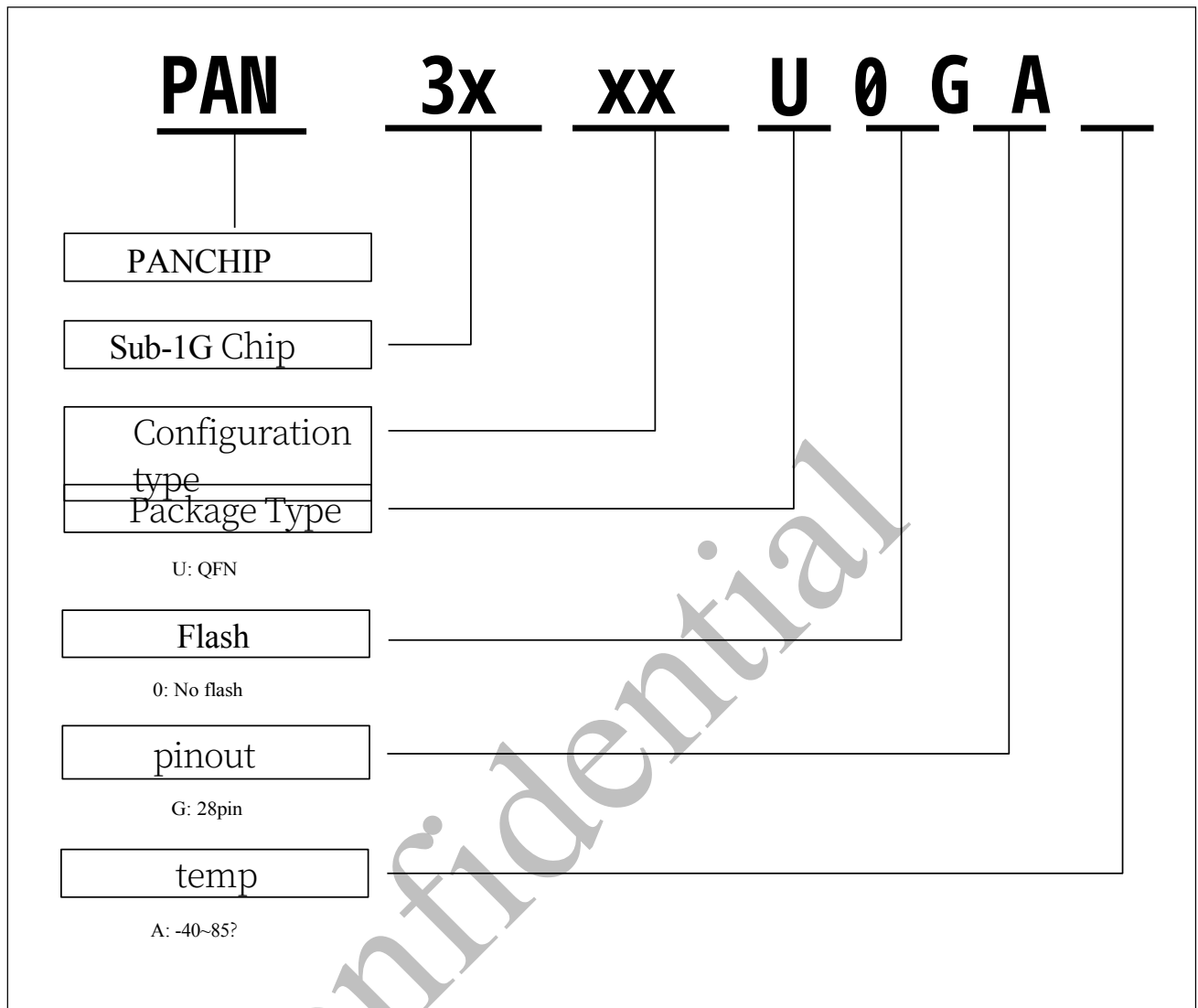
Transceivers table of contents

Table 4-1 Pin Descriptions.....	10
Table 5-1 Absolute Maximum Ratings.....	12
Table 5-2 Voltage and Current (Frequency 490MHz)	12
Table 5-3 Voltage and Current (Frequency 915MHz)	13
Table 5-3 Generalized RF Characteristics (Frequency 490MHz).....	14
Table 5-5 Generalized RF Characteristics (Frequency 915MHz).....	15
Table 9-1 GPIO Port Mode Configuration	32
Table 14-1 QFN28 Package Dimensions	49

catalog of diagrams

Figure 3-1 System Architecture Block Diagram	8
Figure 4-1 Pinouts.....	9
Figure 7-1 PHY Frame Structure	19
Figure 8-1 Single Packet Transmission Mode State Flowchart	21
Figure 8-2 Continuous Transmission Mode Flowchart.....	22
Figure 8-3 Single Packet Receive Mode Flowchart	23
Figure 8-4 Single Packet Receive Mode Flowchart with Timeout	24
Figure 8-5 Continuous Receive Mode Flowchart.....	Figure 8-5 Continuous Receive Mode Flowchart
Figure 9-1 SPI Write Timing.....	29
Figure 9-2 SPI Read Timing.....	29
Figure 9-3 3-Wire SPI Write Timing.....	30
Figure 9-4 3-Wire SPI Read Timing.....	30
Figure 9-5 I2C Start Signal	31
Figure 9-6 I2C Termination Signal	31
Figure 9-7 I2C Write Timing	31
Figure 9-8 I2C Read Timing.....	31
Figure 10-1 Channel Active Detection (CAD).....	35
Figure 10-2 Intelligent SF Identification	38
Figure 11-1 Working Mode Setting Flow	39
Figure 13-1 LDO Mode 433MHz Reference Schematic.....	46
Figure 13-2 DCDC Mode 433MHz Reference Schematic.....	46
Figure 13-3 LDO Mode 490MHz Reference Schematic.....	47
Figure 13-4 DCDC Mode 490MHz Reference Schematic.....	47
Figure 13-5 863MHz Reference Schematic	48
Figure 13-6 902MHz Reference Schematic	48
Figure 14-1 QFN28 Package Diagram.....	49
Figure 15-1 Reflow Soldering Process Profile	50

1 naming convention



2 Ordering Information

Product Model	Chip Type	seal inside	pinout	IO	Supported Bandwidth	SF Factor	bitrate	temp	wrap
PAN3029 U0GA	Sub-1G	QFN	28	4	62.5kHz 125kHz 250kHz 500kHz	SF5~SF12	0.08~ 59.9kbps	-40~85°C	Tape & Reel
PAN3060 U0GA	Sub-1G	QFN	28	4	125kHz 250kHz 500kHz	SF5~SF9	0.5~ 59.9kbps	-40~85°C	Tape & Reel

Before ordering, please consult with sales for the latest mass production information.

Confidential

This manual is described in terms of the largest package and fullest functionality. Please refer to the ordering information on this page for configuration differences between models.

3 System Architecture Block Diagram

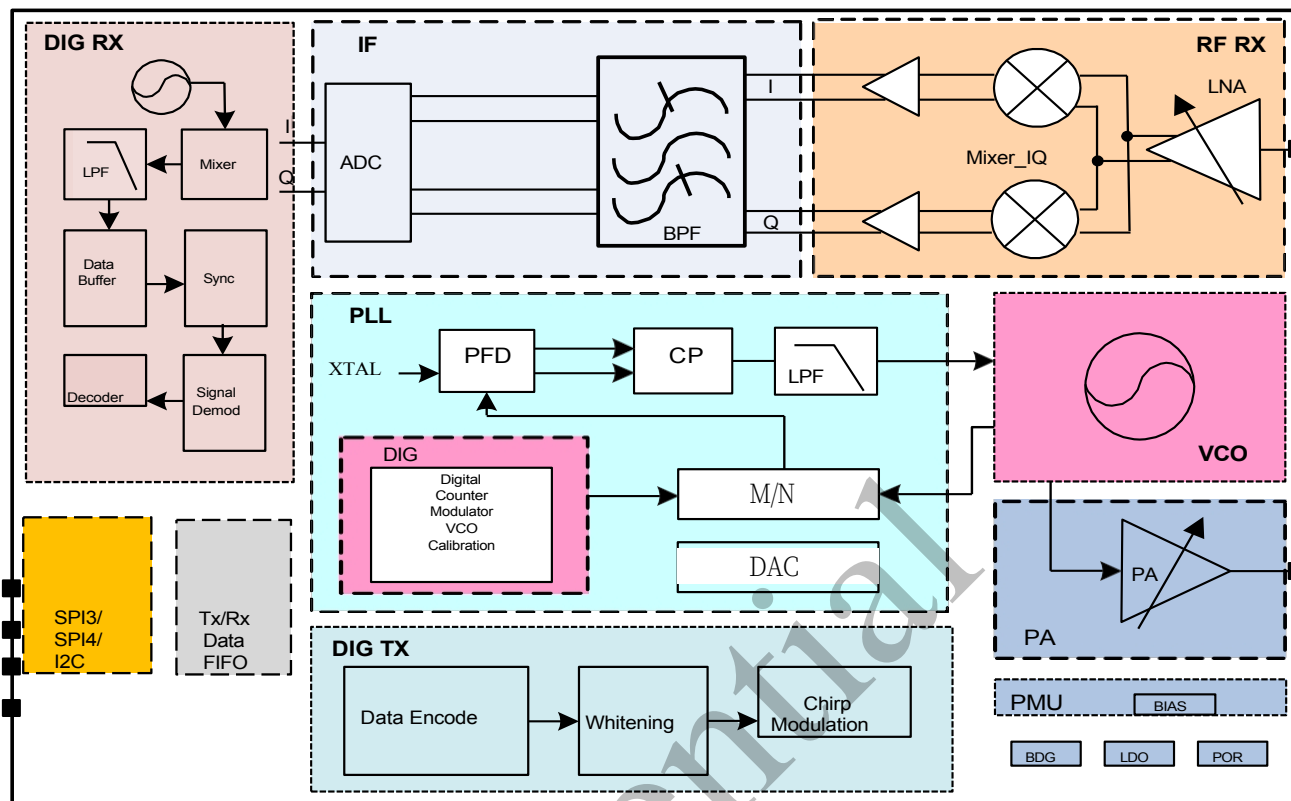


Figure 3-1 System Structure Block Diagram

4 Pin Definitions and Descriptions

4.1 pinout

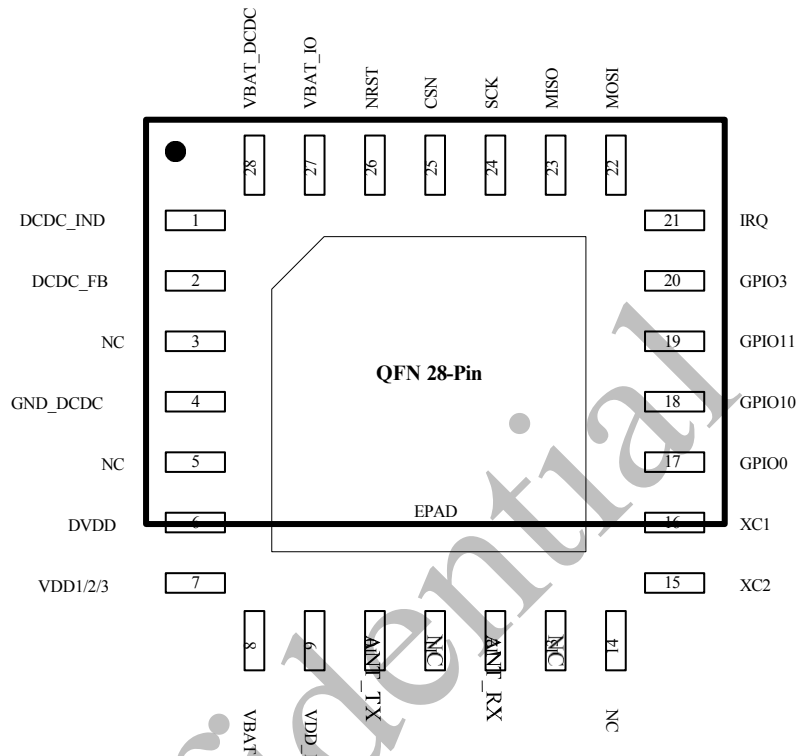


Figure 4-1 Pinout Diagram

4.2 Pin Description

Table 4-1 Pin Descriptions

serial number	notation	typology	functionality
1	DCDC_IND	AO	Internal DCDC output, connected to external series inductor
2	DCDC_FB	AI	Internal DCDC feedback input, DCDC mode connected VDD1/2/3
3	NC		
4	GND_DCDC	G	analogically
5	NC		
6	DVDD	P	Digital Power LDO Outputs
7	VDD1/2/3	P	Analog power supply, DCDC mode to VFB, non-DCDC mode to main power supply
8	VBAT	P	Analog power supply, connected to mains
9	VDD_PA	P	PA Power Supply LDO Output
10	ANT_TX	AO	Transmitter PA Output
11	NC		
12	ANT_RX	AI	Receiver LNA Input
13	NC		
14	NC		
15	XC2	AI	Crystal Input
16	XC1	AO	Crystal Output
17	GPIO0	I/O	Digital IO
18	GPIO10	I/O	Digital IO
		o	External PA control signal
19	GPIO11	I/O	Digital IO
		o	Channel Status Indication Signal
20	GPIO3	I/O	Digital IO
21	IRQ	o	interrupt signal
22	MOSI	I	SPI Data Input Signal
23	MISO	o	SPI Data Output Signal
24	SCK	I	SPI Serial Clock
25	CSN	I	SPI Chip Select Signal
26	NRST	I	reset (a dislocated joint, an electronic device etc)
27	VBAT_IO	P	Digital GPIO Power Supply, Connected to Mains
28	VBAT_DCDC	P	DCDC Power Supply, Connect to Mains



29	E-PAD	G	Transceivers GND pad bottom of chip, needs to be grounded
----	-------	---	---

Confidential

Transceivers

5 Electrical Characteristics

All parameters in this section are based on tests with **TA=25°C** and **supply voltage 3.3V**, unless otherwise specified.

5.1 Absolute maximum rating

Table 5-1 Absolute Maximum Ratings

notation	parameters	conditional	minimum value	typical value	maximum values	unit (of measure)
VDD	vdd1/vdd2/vdd3/vbat/vbat_io	Frequency: 490MHz	-0.3	3.3	3.6	V
VI	Input Voltage		-0.3	-	VDD	V
VO	output voltage		VSS	-	VDD	V
TOP	operating temperature		-40	-	85	°C
TSTG	Storage temperature		-55	-	125	°C

Exceeding operation of the limit values may cause permanent damage to the chip.
Static sensitive equipment, operated in compliance with protection rules.

5.2 DC Characteristics

Test conditions:

- Frequency: 490MHz

Table 5-2 Voltage and Current (Frequency 490MHz)

notation	parameters	minimum value	typical value	maximum values	unit (of measure)	test condition
VDD	power supply	1.8	3.3	3.6	V	TA=25°C, LDO mode
		2	3.3	3.6	V	TA=25°C, DCDC mode
VSS	ground	-	0	-	V	-
IDeepSleep	Deep Sleep Current	-	200	-	nA	-
ITX, 19.5dBm	TX mode operating current	-	91	-	mA	20dBm Output Power
ITX, 18dBm	TX mode operating current	-	76	-	mA	18dBm Output Power
ITX, -2dBm	TX mode operating current	-	13	-	mA	-2dBm output power
IRX, LDO	RX mode operating current	-	8.2	-	mA	Maximum LNA in LDO mode gain (electronics)
IRX, DCDC	RX mode operating current	-	4.1	-	mA	Maximum LNA in DCDC mode gain (electronics)

Transceivers						
VOH	Output High Level Voltage	VDD-0.3	-	VDD	V	-
VOL	Output Low Level Voltage	VSS	-	VSS+0.3	V	-
VIH	Input High Level Voltage	0.8*VDD	-	-	V	-
VIL	Input Low Level Voltage	-	-	0.2*VDD	V	-
SPI_rate	SPI Rate	-	-	10	Mbps	-

Test conditions:

- Frequency: 915MHz

Table 5-3 Voltage and Current (Frequency 915MHz)

notation	parameters	minimum value	typical value	maximum values	unit (of measure)	test condition
VDD	power supply	1.8	3.3	3.6	V	TA=25°C, LDO mode
		2	3.3	3.6	V	TA=25°C, DCDC mode
VSS	structural particle: used before a verb or adjective, linking it preceding the verb or adjective	-	0	-	V	-
IDeepSleep	Deep Sleep Current	-	200	-	nA	-
ITX, 19.5dBm	TX mode operating current	-	107	-	mA	20dBm Output Power
ITX, 18dBm	TX mode operating current	-	103	-	mA	18dBm Output Power
ITX, -2dBm	TX mode operating current	-	16	-	mA	-2dBm output power
IRX, LDO	RX mode operating current	-	9.1	-	mA	Maximum LNA in LDO mode gain (electronics)
IRX, DCDC	RX mode operating current	-	4.7	-	mA	Maximum LNA in DCDC mode gain (electronics)
VOH	Output High Level Voltage	VDD-0.3	-	VDD	V	-
VOL	Output Low Level Voltage	VSS	-	VSS+0.3	V	-
VIH	Input High Level Voltage	0.8*VDD	-	-	V	-
VIL	Input Low Level Voltage	-	-	0.2*VDD	V	-
SPI_rate	SPI Rate	-	-	10	Mbps	-

Transceivers
5.3 RF Characteristics

Test conditions:

- Frequency: 490MHz
- Error correction code= 4/8
- False packet rate≤ 5%
- Payload length= 10Bytes

Table 5-4 Generalized RF Characteristics (Frequency 490MHz)

notation	descriptive	conditional	minimal	typical case	greatest	unit (of measure)
common frequency						
F _{op}	operating frequency	-	408	-	565	MHz
		-	816	-	1080	MHz
F _{xtal}	crystal frequency	-	-	32	-	MHz
R _S	Crystal series resistance	-	-	30	50	Ω
C _{FOOT}	Crystal External Capacitance	-	8	15	22	pF
C _{LOAD}	Crystal Load Capacitance	-	6	10	12	pF
F _{TOL}	Initial frequency tolerance	-	-	±10	-	ppm
BR	Bit Rate (PAN3029 Series)	-	0.08	-	59.9	kbps
	Bit Rate (PAN3060 Series)	-	0.5	-	59.9	kbps
emitters						
PLP _{WAN}	output power	-	-30	19	20	dBm
refraction						
RF_62.5 (LDO)	RF sensitivity, long range mode, highest LNA gain, 62.5 kHz bandwidth using separate RX/TX channels	SF= 7	-	-129	-	dBm
		SF= 10	-	-137	-	
		SF= 12	-	-143	-	
RF_125 (LDO)	RF sensitivity, long range mode, highest LNA gain, 125 kHz bandwidth using separate RX/TX channels	SF= 7	-	-126	-	dBm
		SF= 10	-	-134	-	
		SF= 12	-	-140	-	
RF_250 (LDO)	RF sensitivity, long range mode, highest LNA gain, 250 kHz bandwidth using separate RX/TX channels	SF= 7	-	-124	-	dBm
		SF= 10	-	-132	-	
		SF= 12	-	-137	-	
RF_500 (LDO)	RF sensitivity, long range mode, highest LNA gain, 500 kHz bandwidth using separate RX/TX channels	SF= 7	-	-121	-	dBm
		SF= 10	-	-129	-	
		SF= 12	-	-134	-	
RF_62.5 (DCDC)	RF sensitivity, long range mode, highest LNA gain, 62.5 kHz bandwidth using separate RX/TX channels	SF= 7	-	-125	-	dBm
		SF= 10	-	-133	-	
		SF= 12	-	-139	-	
RF_125 (DCDC)	RF sensitivity, long range mode, highest LNA gain, 125 kHz bandwidth using separate RX/TX channels	SF= 7	-	-122	-	dBm
		SF= 10	-	-130	-	
		SF= 12	-	-137	-	
RF_250 (DCDC)	RF sensitivity, long range mode, highest LNA gain, 250 kHz bandwidth using separate RX/TX channels	SF= 7	-	-120	-	dBm
		SF= 10	-	-128	-	
		SF= 12	-	-134	-	

Transceivers						
RF_500 (DCDC)	RF sensitivity, long range mode, highest LNA gain, 500 kHz bandwidth using separate RX/TX channels	SF= 7 SF= 10 SF= 12	- - -	-117 -125 -130	- - -	dBm

Test conditions:

- Frequency: 915MHz
- Error correction code= 4/8
- False packet rate≤ 5%
- Payload length= 10Bytes

Table 5-5 Generalized RF Characteristics (Frequency 915MHz)

notation	descriptive	conditional	minimal	typical case	greatest	unit (of measure)
common frequency						
Fop	operating frequency	-	408	-	565	MHz
		-	816	-	1080	MHz
Fxtal	crystal frequency	-	-	32	-	MHz
RS	Crystal series resistance	-	-	30	50	Ω
CFOOT	Crystal External Capacitance	-	8	15	22	pF
CLOAD	Crystal Load Capacitance	-	6	10	12	pF
FTOL	Initial frequency tolerance	-	-	±10	-	ppm
BR	Bit Rate (PAN3029 Series)	-	0.08	-	59.9	kbps
	Bit Rate (PAN3060 Series)	-	0.5	-	59.9	kbps
emitters						
PLPWAN	output power	-	-30	19	20	dBm
refraction						
RF_62.5 (LDO)	RF sensitivity, long range mode, highest LNA gain, 62.5 kHz bandwidth using separate RX/TX channels	SF= 7	-	-128	-	dBm
		SF= 10	-	-136	-	
		SF= 12	-	-141	-	
RF_125 (LDO)	RF sensitivity, long range mode, highest LNA gain, 125 kHz bandwidth using separate RX/TX channels	SF= 7	-	-125	-	dBm
		SF= 10	-	-132	-	
		SF= 12	-	-138	-	
RF_250 (LDO)	RF sensitivity, long range mode, highest LNA gain, 250 kHz bandwidth using separate RX/TX channels	SF= 7	-	-121	-	dBm
		SF= 10	-	-129	-	
		SF= 12	-	-135	-	
RF_500 (LDO)	RF sensitivity, long range mode, highest LNA gain, 500 kHz bandwidth using separate RX/TX channels	SF= 7	-	-119	-	dBm
		SF= 10	-	-127	-	
		SF= 12	-	-132	-	
RF_62.5 (DCDC)	RF sensitivity, long range mode, highest LNA gain, 62.5 kHz bandwidth using separate RX/TX channels	SF= 7	-	-128	-	dBm
		SF= 10	-	-136	-	
		SF= 12	-	-140	-	
RF_125 (DCDC)	RF sensitivity, long range mode, highest LNA gain, 125 kHz bandwidth using separate RX/TX channels	SF= 7	-	-125	-	dBm
		SF= 10	-	-132	-	
		SF= 12	-	-137	-	

RF_250	RF Sensitivity, Long Range Mode, Maximum LNA Gain.	SF= 7	-	-120	-	dBm
(DCDC)	250 kHz bandwidth using separate RX/TX channels	SF= 10 SF= 12	- -	-129 -135	- -	
RF_500	RF sensitivity, long range mode, highest LNA gain, 500 kHz bandwidth using separate RX/TX channels	SF= 7 SF= 10 SF= 12	- - -	-118 -126 -130	- - -	dBm
(DCDC)						

Confidential

6 Frequency synthesizer circuit

The chip integrates high-performance RF transceiver circuits and can support multiple Sub-1GHz frequency bands such as 400M and 800M.

The chip contains a high performance, low phase noise, fully integrated fractional frequency synthesizer. The frequency synthesizer is used in receive mode to generate the LO frequency used by the downconverting mixer. In transmit mode, it is also used to directly generate the modulated RF carrier. The Fractional Frequency Synthesizer provides excellent phase noise performance, frequency resolution of less than 100Hz, and low power consumption. The frequency synthesizer has a fast frequency stabilization function that enables receive and transmit frequency stabilization to be completed in a very short time to optimize chip power consumption.

Confidential

7 Modem

The chip utilizes a modem that supports ChirpIoT™ transmissions to achieve ultra-high communication link budgets through superior spread spectrum modulation and forward error correction. This increases the range and stability of the radio communication link compared to conventional FSK-based modulation.

The ChirpIoT™ modem has been greatly enhanced in terms of anti-interference capability. Compared with FSK modulation, the ChirpIoT™ modem is able to inhibit co-channel interference by up to 19dB, i.e., the energy of the interfering signals can be up to 19dB higher than that of the ChirpIoT™ signals, which greatly ensures that the ChirpIoT™ signals can be simply coexisted with other modulation systems and are used in a large number of hybrid network communication applications. This greatly ensures that ChirpIoT™ signals can simply coexist with other modulation systems and are used extensively in hybrid network communications.

7.1 modulation parameter

The chip optimizes the transmission performance of ChirpIoT™ signals by configuring different modulation parameters to achieve a balance of transmission rate, transmission distance, and transmission reliability. The specific parameters are listed below:

- Signal Bandwidth (BW)
- Spreading factor (SF)
- Code Rate (CR)
- Low rate mode (LDR)

7.1.1 signal bandwidth

The PAN3029 supports four signal bandwidths, 500kHz, 250kHz, 125kHz, and 62.5kHz. The PAN3060 supports three signal bandwidths, 500kHz, 250kHz, and 125kHz.

Different signal bandwidths can be selected through register configuration.

7.1.2 spreading factor

The PAN3029 supports 8 different spreading factors from SF5 to SF12 and the PAN3060 supports SF5 to SF9.

There are 5 different spreading factors. The larger the spreading factor, the lower the data transmission net load rate.

7.1.3 coding rate

The chip adopts forward error correction technology and supports four different code rates: CR45, CR46, CR47 and CR48. This forward error correction technology can correct a certain range of error bits to improve transmission reliability.

7.1.4 low speed mode

Low-rate mode is a communication mode of the ChirpIoT™ modulation technology that reduces the number of effective bits carried per unit of time by reducing the number of bits carried per unit of time.

information, improving the high reliability of signal transmission.

7.2 frame structure

The frame structure of the ChipIoT™ transmission signal consists of four parts, Signal Preamble lead in segment, Header information segment, Payload data segment, and CRC checksum segment, as shown below.

Preamble	Header	CRC	Payload	Payload CRC
----------	--------	-----	---------	-------------

Figure 7-1 PHY Frame Structure

Confidential

8 MAC Design

8.1 SEND RECEIVE MODE

The chip's transmitter has two unique modes of operation and the receiver has three unique modes of operation. Transmitter operating modes:

- single-packet delivery mode
- Continuous

Transmit Mode The operating mode of the receiver:

- Single packet reception mode
- Single packet receive mode with timeout
- Continuous reception mode

8.2 transmitter mode

In MAC transmit mode, there are two modes: single packet transmit and continuous transmit mode. The mode selection register can be used to select the mode. After the TX mode is selected in the mode selection register, the state machine enters the TX preparation state and waits for the FIFO to be filled with data before entering the TX transmission state.

When the MAC enters the TX transmit mode, the power consumption control module turns on the analog module circuits such as LDO module, PLL module, PA, etc., and then sends the TX start signal to the Modem module to start transmitting data.

After the data is sent, turn off the analog module circuits such as LDO module, PLL module, PA, and so on. Then send the IRQ signal to the MCU. After the MCU clears the IRQ, it can configure the REG_OPT_MODE register and enter the STB3 mode to end the TX process.

8.2.1 single-packet delivery mode

The state flow diagram for the single packet transmit mode is shown in Figure 8-1.

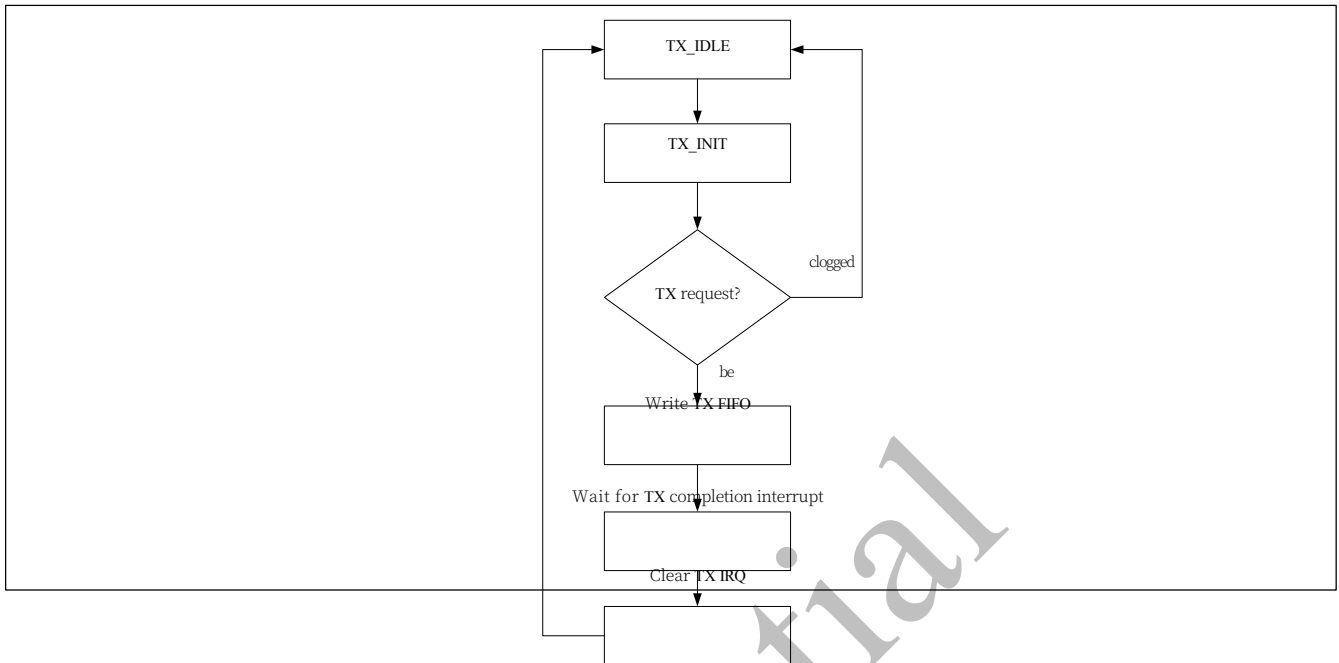


Figure 8-1 Single-Packet Transmission Mode State Flowchart

8.2.2 Continuous transmission mode

The state flowchart for continuous transmit mode is shown in Figure 8-2.

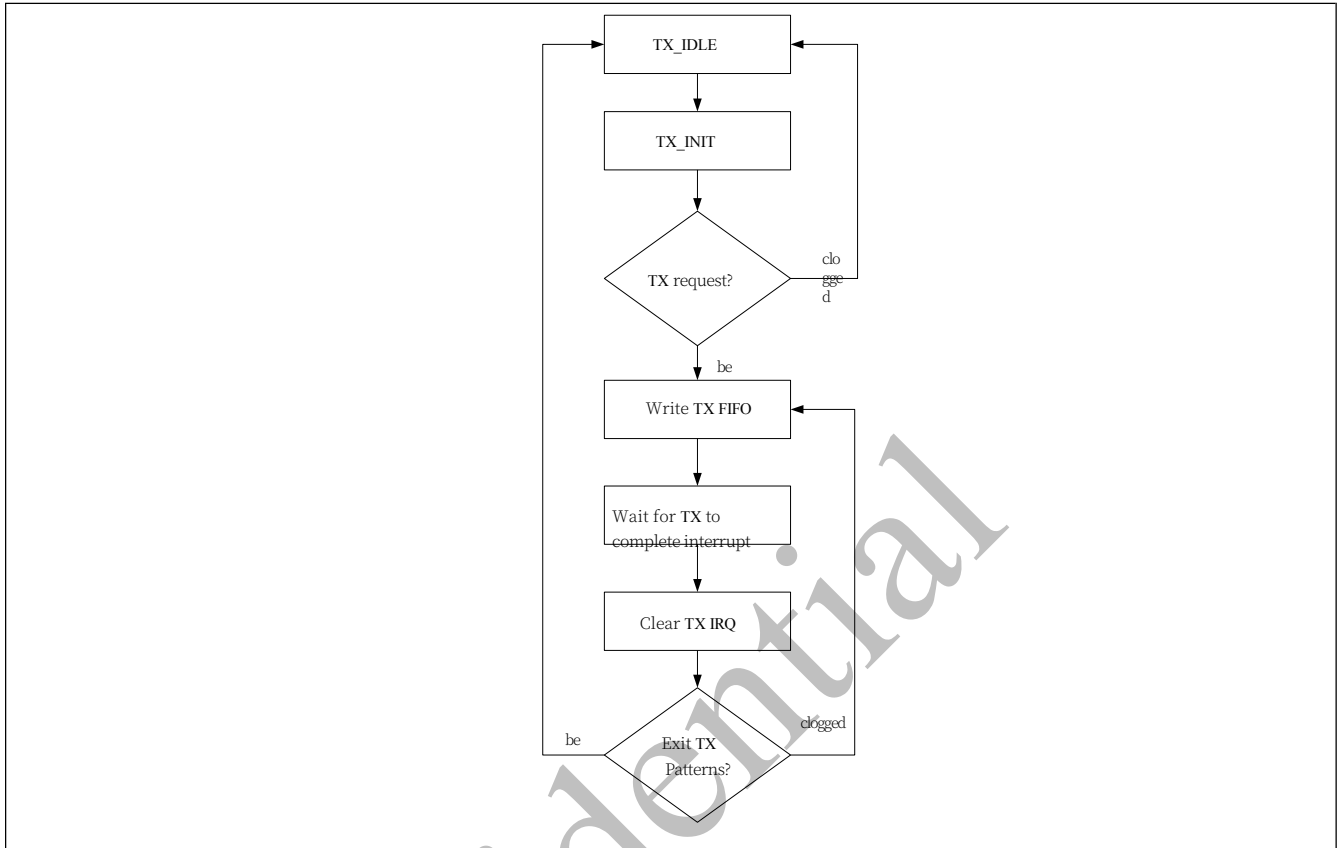


Figure 8-2 Continuous Transmission Mode Flowchart

8.3 receiver mode

Among the MAC receive modes, there are single packet receive, single packet receive with timeout, and continuous receive modes, which are selected with the 2-bit mode select register.

The state machine enters the RX receive state after the RX mode is selected in the mode selection register.

When the MAC enters the receive mode, the power control module turns on the analog module circuits such as LDO, PLL, etc., and sends the

RX start signal to Modem to start receiving data.

After receiving the data, turn off the analog module circuits such as LDO and PLL in turn. Then send the IRQ signal to the MCU. After the MCU clears the IRQ, it can configure the REG_OPT_MODE register and enter the STB3 mode to finish the RX process.

8.3.1 Single packet reception mode

The state flow diagram for the single packet receive mode is shown in Figure 8-3.

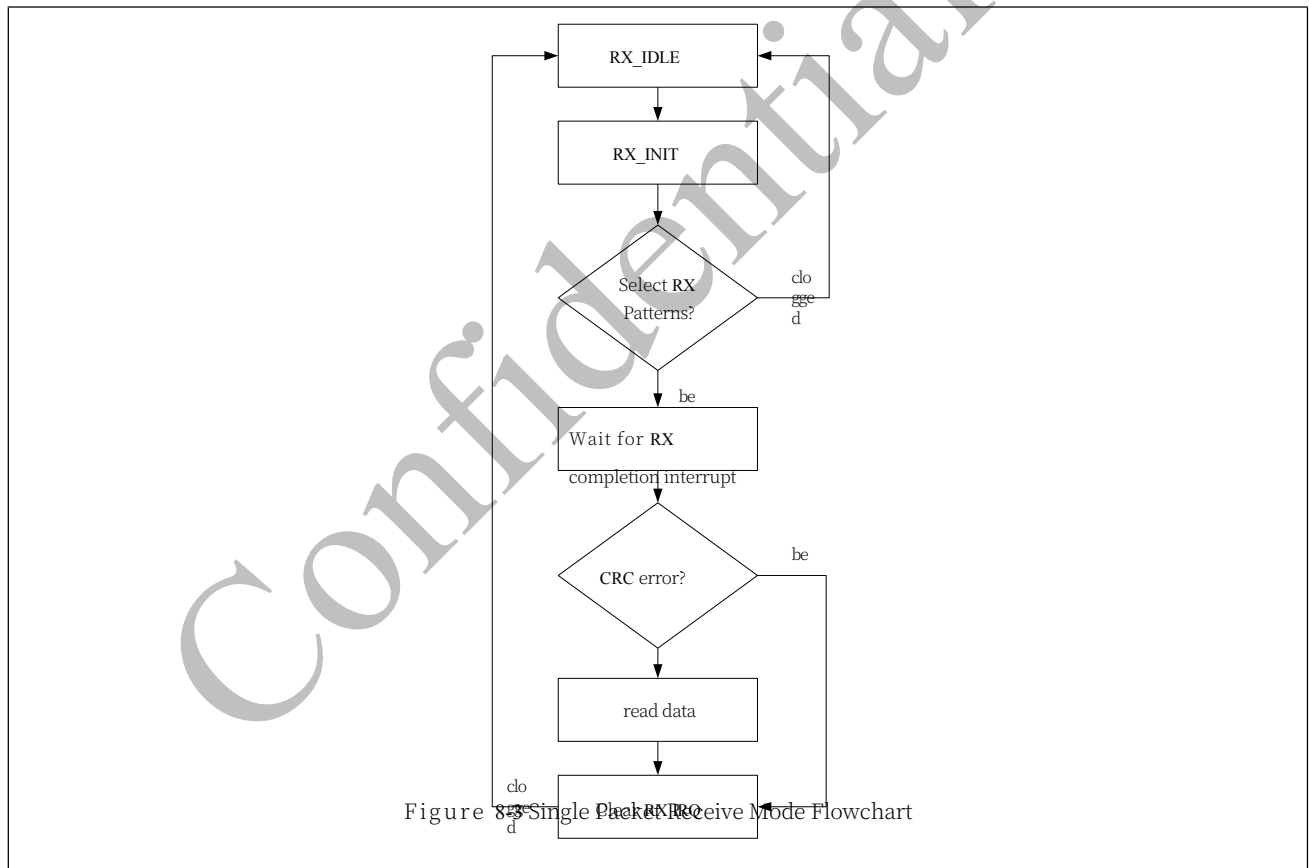


Figure 8-3 Single Packet Receive Mode Flowchart

Transceivers

8.3.2 Single packet receive mode with timeout

The state flow diagram for the single packet receive mode with timeout is shown in Figure 8-4.

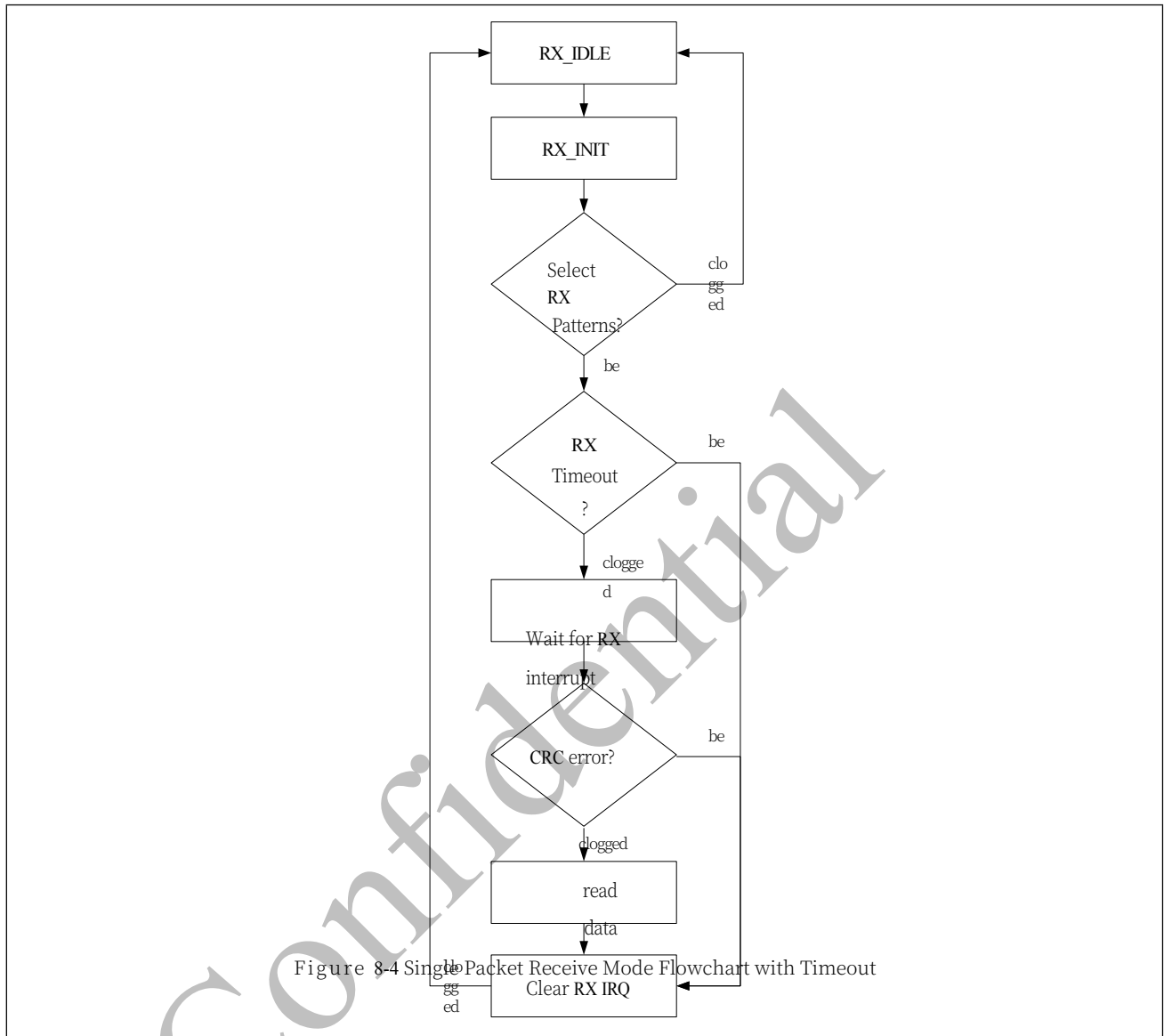


Figure 8-4 Single Packet Receive Mode Flowchart with Timeout

8.3.3 Continuous reception mode

The state flow diagram for the continuous reception mode is shown in Figure 8-5.

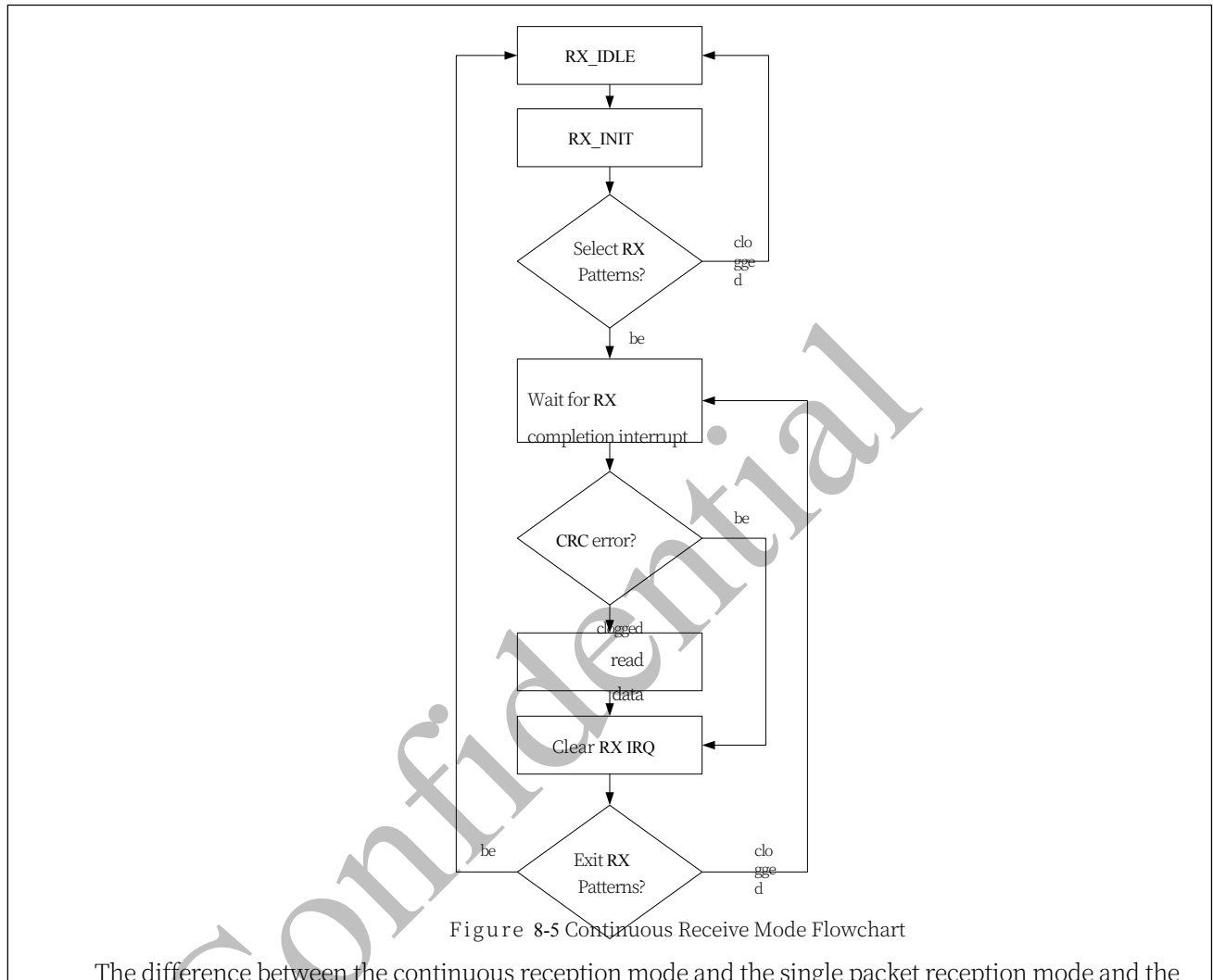


Figure 8-5 Continuous Receive Mode Flowchart

The difference between the continuous reception mode and the single packet reception mode and the single packet reception mode with timeout is that after the data is received, an IRQ signal is sent to the MCU, and the MCU clears the IRQ to start the next data reception. To exit the continuous reception mode, select the crystal output mode in the mode selection register, and then turn off the analog module circuits, such as LDO module and PLL module, to end the process of continuous reception mode once.

8.4 IRQ Interrupt

All interrupts on the chip share an external interrupt IO pin, **IRQ**, which is active high. There are 7 types of interrupts as follows. The interrupts can be turned on through the interrupt enable register and the interrupt status can be read through the interrupt status register.

Interrupt Enable Register	descriptive
REG_MAPM_IRQ_MASK	IRQ MASK enable in MAPM mode 0: Generate IRQ interrupt for MAPM 1: IRQ interrupt without MAPM generation
REG_RX_HMP_ERR_MASK	Error Correction Code Error IRQ Interrupt Mask 0: IRQ interrupts without error correction code errors 1: IRQ interrupts that generate error correction code errors
REG_RX_PLHD_DONE_MASK	Early Interrupt Mode Mask 0: No early interrupt mode IRQ interrupts are generated 1: Generate IRQ interrupts for early interrupt mode
REG_RX_DONE_MASK	rx complete irq mask 0: Do not generate rx-complete IRQ interrupt 1: Generate IRQ interrupt for rx completion
REG_RX_PL_CRC_ERR_MASK	Rx payload crc error irq mask 0: IRQ interrupt without Rx payload crc error 1: IRQ interrupt generating Rx payload crc error
REG_RX_TIMEOUT_MASK	Rx timeout irq mask 0: IRQ interrupt without Rx timeout 1: IRQ interrupt that generates Rx timeout
REG_TX_DONE_MASK	Tx completion irq mask 0: IRQ interrupt for Tx completion is not generated 1: Generate IRQ interrupt for Tx completion

Interrupt Status Register	descriptive
REG_MDM_MAPM_IRQ	IRQ interrupt for MAPM
REG_RX_HM_P_ERR_IRQ	IRQ interrupt for error correction code errors
REG_RX_PLHD_DONE_IRQ	IRQ interrupts in early interrupt mode
REG_RX_DONE_IRQ	IRQ interrupt completed by Rx
REG_RX_PL_CRC_ERR_IRQ	Rx payload crc wrong IRQ interrupt
REG_RX_TIMEOUT_IRQ	IRQ interrupt with Rx timeout
REG_TX_DONE_IRQ	Tx Completed IRQ Interrupt

8.5 CAD-IRQ Interrupt

The CAD-IRQ interrupt pin of the chip multiplexes the GPIO11 pin, and it is necessary to configure the GPIO11 hardware into CAD-IRQ output mode when using the chip.

Register Name	register location	descriptive
REG_OUT_FUNC_SEL_CAD	page0, 0x5e, bit6	CAD Function Selection When REG_OUT_FUNC_SEL_CAD is 1, the GPIO is a normal IO, whose output can be configured by register PAD_OUT_REG; When REG_OUT_FUNC_SEL_CAD is 0, the GPIO outputs are CAD Function Pins.
PAD_OUT_REG_CAD	page0, 0x68, bit3	Registers that control the output of GPIO11.

Confidential

9 interface design

The chip can support 4-wire SPI, SPI and I2C to registers, send/receive FIFOs, 4-wire SPI and I2C at the same time by default, and 3-wire SPI needs to be configured through the register SPI3_EN_REG to take effect.

Note: The FIFO address is a special area, the register address of this area is the entry address of these special areas, and the operation mode is different from that of the regular registers. please refer to section 9.4 FIFO for the FIFO operation mode.

9.1 Four-wire SPI

The chip implements a Slave of the SPI bus for reading and writing registers and FIFOs. The SPI bus is a four-wire system, respectively:

- SCK (clock)
- CSN (chip select signal, active low)
- MOSI (data input)
- MISO (data output)

Among them, SCK, CSN, MOSI are controlled by the host Master, and MISO is controlled by the Slave. The communication process starts with CSN level low and ends with CSN level high. The host master sends data through MOSI, and MISO receives data, data is generated on the falling edge of SCK, and data is sampled on the rising edge.

The information transmitted by the Master consists of two parts: Address Byte and Data Byte. The first 7 bits of Address Byte are address bits *addr*; the last 1 bit is read/write bit *wr*, which is set to 1 for write operation and 0 for read operation. SPI has three transmission modes:

- **Single:** Single byte transmission mode. The information is only 2byte, Master sends Address Byte through MOSI, if it is write operation, Master continues to send Data Byte through MOSI; if it is read operation, Master reads Data Byte replied by Slave on MISO.
- **Burst:** Burst continuous transmission mode. The message is larger than 2 bytes, Address Byte is followed by a number of Data Bytes, there is no need to add Address Bytes between Data Bytes, the slave will automatically increment the address between each Data Byte internally. The CSN signal will be pulled high after the last Data Byte, and remain low during the rest of the message transmission.
- **FIFO:** FIFO read/write mode. In this mode, single byte or continuous transmission can be realized, and the transmission rules are the same as Single mode and Burst mode, the difference is that the address bit *addr* in Address Byte can only be configured as 0x01, and the Slave does not do address increment operation between Data Bytes.

In addition, the MISO can be to input high resistance state by configuring REG_OUT_PAD_MODE register to realize the 4-wire SPI one-master-multi-slave connection architecture.

Transceivers

9.1.1 Four-wire SPI Write Timing

The SPI write timing is as follows:

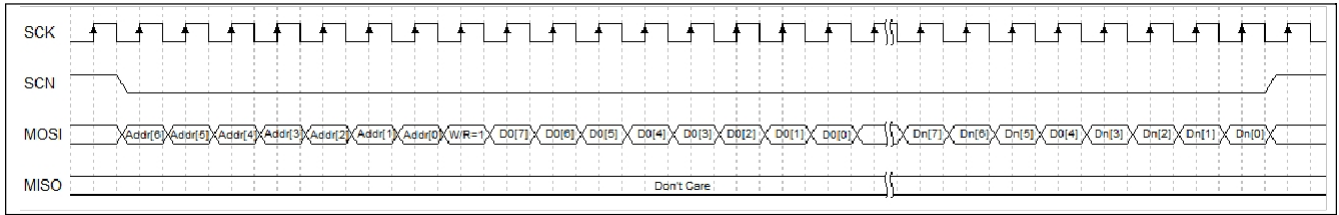


Figure 9-1 SPI Write Timing

9.1.2 Four-wire SPI read timing

The SPI read timing is as follows:

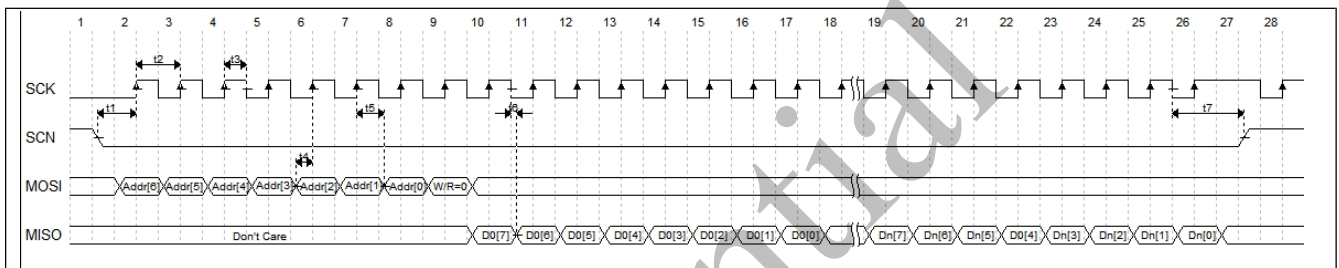


Figure 9-2 SPI Read Timing

9.1.3 Four-wire SPI Timing Requirements

All SPI timing requirements in the following table are listed below:

notation	descriptive	minimum value	typical value	maximum values	unit (of measure)
t1	SCN falling edge to SCK rising edge	32	-	-	ns
t2	SCK Clock Period	100	-	-	ns
t3	SCK high time	50	-	-	ns
t4	MOSI to SCK clock SETUP time	7	-	-	ns
t5	MOSI to SCK clock HOLD time	7	-	-	ns
t6	SCK falling edge to MISO delay time	0	-	15	ns
t7	SCK to SCN rising edge time	0	-	150	ns

9.2 Three-wire SPI

The chip supports 4-wire SPI and I2C by default, if you need to use 3-wire SPI, you need to configure register SPI3_EN_REG after power-on to make 3-wire SPI effective. The three-wire SPI signals include:

- SCK (clock)
- CSN (chip select signal, active low)
- MOSI (data input/output)

9.2.1 Three-wire SPI Write Timing

The SPI write timing is as follows:

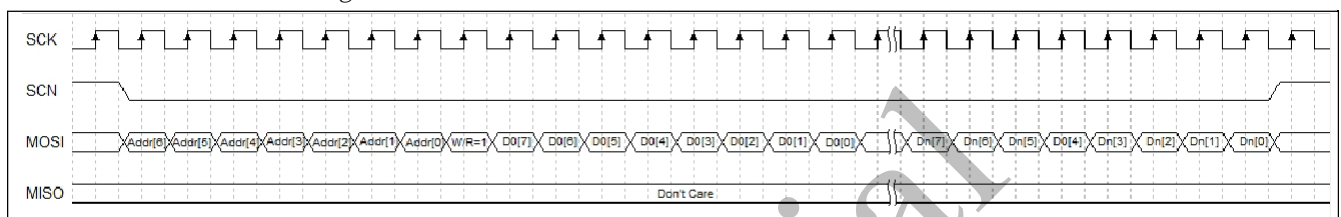


Figure 9-3 3-Wire SPI Write Timing

9.2.2 Three-wire SPI read timing

The SPI read timing is as follows:

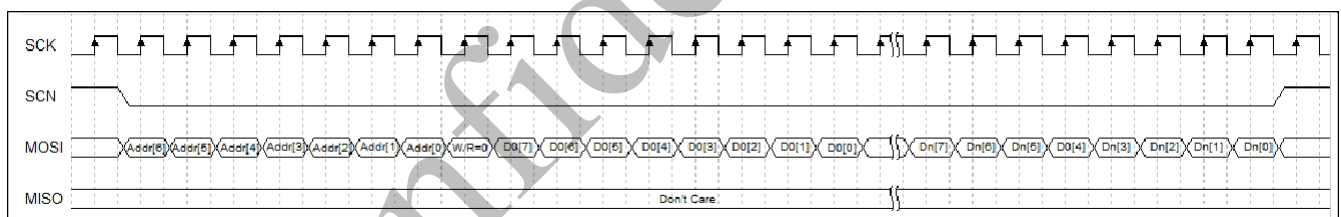


Figure 9-4 3-Wire SPI Read Timing

9.2.3 Three-wire SPI Timing Requirements

Please refer to section 9.1.3.

9.3 I2C

The chip supports I2C by default, and the I2C signals SCL and SDA multiplex the SCK and MOSI signals of SPI.

Note: When using I2C, the CSN signal is required to be high.

The device address bit width of I2C is 7bit and the content is 0x72. The timing of I2C start signal Start is shown below:

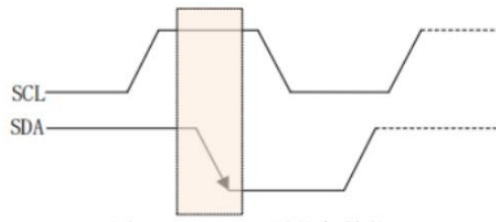


Figure 9-5 I2C Start Signal

The timing diagram for the I2C termination signal Stop is shown below:



Figure 9-6 I2C Termination Signal

9.3.1 I2C Write Timing

The I2C write timing is as follows:

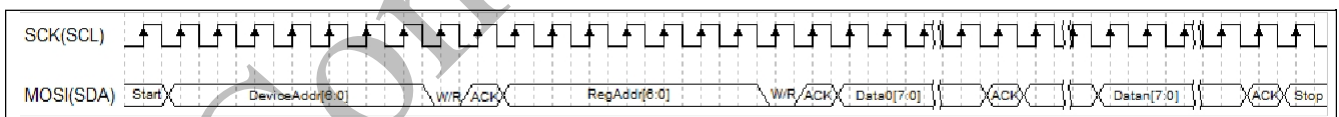


Figure 9-7 I2C Write Timing

9.3.2 I2C Read Timing

The I2C read timing is as follows:

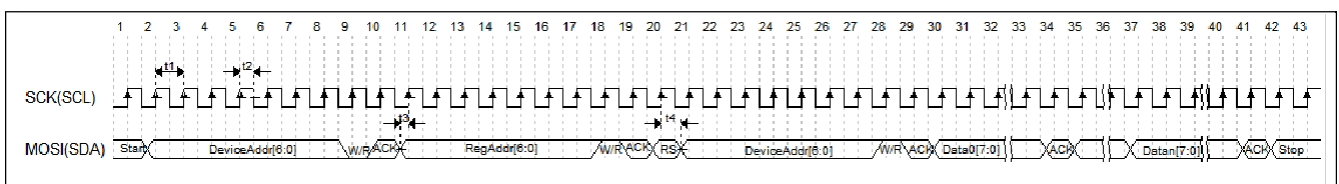


Figure 9-8 I2C Read Timing

9.3.3 I2C Timing Requirements

notation	descriptive	minimum value	typical value	maximum values	unit (of measure)
t1	SCL Clock Period	200	-	-	ns
t2	SCL high level time	100	-	-	ns
t3	SDA to SCL clock SETUP time	12	-	-	ns
t4	SDA to SCL clock HOLD time	12	-	-	ns

9.4 FIFO

The chip has a 256bytes FIFO to store TX module transmit data and RX module decode data. FIFO of single-port RAM, which can only realize the storage and reading of single packet data information. In case a packet of data is already stored in the FIFO, it should be written after reading this packet, otherwise the previous packet of data in the FIFO will be overwritten. FIFO can be read and written by Modem, SPI and I2C in STB3 and later operating modes. The read/write address of FIFO is 0x01, through which SPI or I2C can read/write the contents of FIFO.

9.5 GPIO

The chip supports multiple GPIO inputs and outputs, and the GPIOs can be controlled by register configuration, please refer to section 12.4 for register configuration. Each GPIO port can be configured as internal pull up/pull down input, floating input, push-pull output, etc. The configuration is shown in Table 9-1.

Table 9-1 GPIO Port Mode Configuration

DIEN	OE	OUT	PUEEN	PDEEN	IO Direction	IO Status
0	1	0/1	0	0	push-pull output	0/1
0	0	x	0	0	open-drain output	high resistance
1	0	x	0	0	importation	high resistance
1	0	x	1	0	importation	pull up
1	1	x	0	1	importation	pull down

9.5.1 External PA Control GPIO

GPIO10 has two configurable output modes: common IO and PA control pins. Through the register REG_OUT_FUNC_SEL_PA configures different modes of GPIO10.

GPIO10 is configured as a normal IO, it can output high and low levels like other GPIOs.

When GPIO10 is configured for PA control and the external PA mode is configured to be on through the registers, GPIO10 can be used as an enable control pin for the external PA, i.e., GPIO10 outputs a high level only when transmitting data in the transmit mode, and outputs a low level at other times.

Register Name	register location	descriptive
REG_OUT_FUNC_SEL_PA	page0, 0x5e, bit7	GPIO10 Function Selection When REG_OUT_FUNC_SEL_PA is 1, GPIO10 is a normal IO, its output can be configured by register PAD_OUT_REG_PA; when REG_OUT_FUNC_SEL_PA is 0, GPIO10 output is external. Set the PA function pin.
PAD_OUT_REG_PA	page0, 0x68, bit2	Registers that control the output of GPIO10. When REG_OUT_FUNC_SEL_PA is 1, GPIO10 is a normal IO, whose output can be configured by register PAD_OUT_REG_PA.

9.5.2 CAD Monitoring GPIO

GPIO11 has two configuration modes: Common IO and CAD to detect pins. Through the register

REG_OUT_FUNC_SEL_CAD Configures different modes of GPIO11.

GPIO11 is configured as a normal IO, it can output high and low levels, no different from other GPIOs. When GPIO11 is configured as a CAD signal detection interrupt output, and the CAD function is also configured to be on via registers, and the chip is in the RX state, when a signal with the same bandwidth and spreading factor parameters as those currently configured is received, GPIO11 generates a continuous and stable high level indicating that the chip searches for an over-the-air signal, which is not pulled down until the signal disappears; in all other cases and modes, GPIO11 outputs a low level; in all other cases and modes, GPIO11 outputs a low level. This high level will not be pulled down until the signal disappears; in other cases and modes, GPIO11 outputs a low level. Otherwise, GPIO11 outputs a low level. For details of CAD configuration, please refer to section 9.4.

The CAD-IRQ interrupt pin of the chip multiplexes the GPIO11 pin, and it is necessary to configure the GPIO11 hardware into CAD-IRQ output mode when using the chip.

Register Name	register location	descriptive
REG_OUT_FUNC_SEL_CAD	page0, 0x5e, bit6	CAD Function Selection When REG_OUT_FUNC_SEL_CAD is 1, the GPIO is a normal IO, whose output can be configured by register PAD_OUT_REG; When REG_OUT_FUNC_SEL_CAD is 0, the GPIO outputs are CAD Function Pins.
PAD_OUT_REG_CAD	page0, 0x68, bit3	Registers that control the output of GPIO11. When REG_OUT_FUNC_SEL_CAD is 1, GPIO11 is a common IO whose output can be configured by register PAD_OUT_REG_CAD.

10 ChirpIoT Special Feature Description

10.1 RSSI Functions

10.1.1 RSSI indication of ChirpIoT signals

The chip supports signal energy statistics function. When the chip receives ChirpIoT data, the internal logic of the chip will count the RSSI value of the signal strength of the current packet, but after the packet reception is finished, the MCU can read the register `RSSI_R32_HOLD` or `RSSI_RBW_HOLD` through the SPI interface to get the RSSI value of the current packet.

10.1.2 Channel Detection RSSI Indication

The RSSI function also supports the indication of energy statistics when detecting channels. When the chip enters the RX state, it can perform RSSI energy counting once through the configuration register `REG_RSSI_PRE_RD`, and then it can retrieve the current signal RSSI energy through the register `RSSI_R32_HOLD`.

`REG_RSSI_PRE_RD` is configured first to 0 and then to 1 to update the RSSI value once, with the following register locations:

Register Name	register location	descriptive
<code>REG_RSSI_PRE_RD</code>	page2, 0x06, bit2	Detects the channel RSSI read switch and updates the RSSI value once when configured from 0 to 1.

10.2 Signal Noise Energy Statistics Function

The chip supports the signal noise statistics function. When the chip receives ChirpIoT data, the internal logic of the chip will calculate the signal energy and noise energy of the current packet, but after the packet reception is completed, the MCU can read the signal energy value of register `SIG_POW_AVG` and the noise energy value of register `NOI_POW_AVG` through the SPI interface, and then the MCU can calculate the standard formula to get the packet's signal-to-noise ratio of the packet.

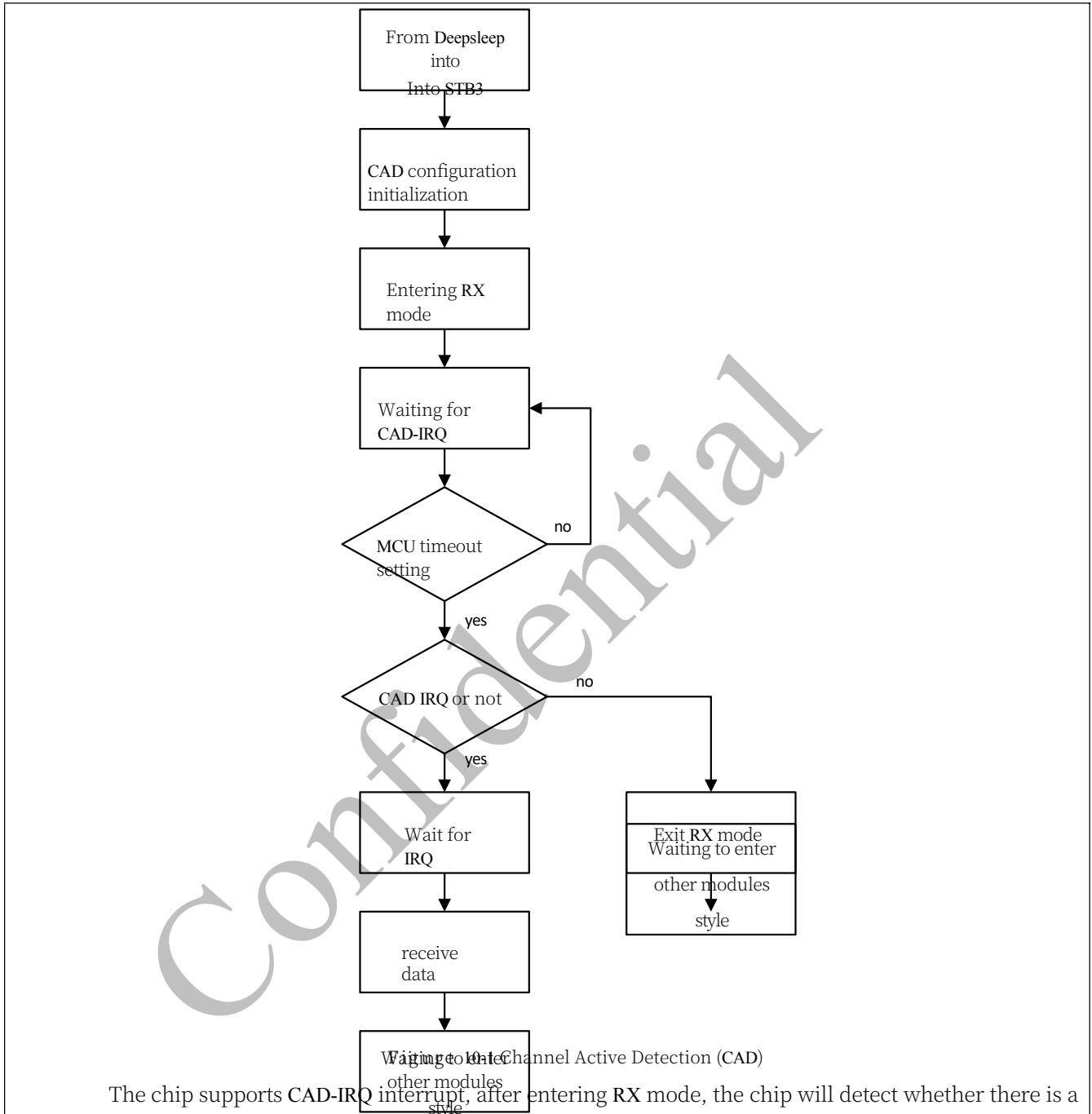
10.3 Channel Active Detection (CAD)

Since the ChirpIoT™ signal has excellent reception performance and can still ensure normal signal reception and demodulation even when it is lower than the receiver noise floor, in such a case, it is no longer sufficient to use the signal strength indicator RSSI to determine whether the channel is occupied or not. Therefore, the chip is specially designed with a channel activity detector to detect the ChirpIoT™ signal to ensure that the channel is occupied by ChirpIoT™ signals from other devices even when it is lower than the receiver noise floor. Therefore, the chip is designed with a channel activity detector to detect ChirpIoT™ signals to ensure that the channel is occupied by ChirpIoT™ signals from other devices even when the noise floor of the receiver is lower than that of the receiver.

The detection function of the Channel Interaction Detector is divided into two parts, one for ChirpIoT™ Signal Pilot Segment Detection and one for ChirpIoT™ Signal Data Segment Detection.

10.3.1 ChirpIoT signaling lead segment detection

As shown in Figure 10-1:



The chip supports CAD-IRQ interrupt, after entering RX mode, the chip will detect whether there is a ChirpIoT™ pilot signal in the channel, and if there is, it will set the CAD-IRQ high, the external MCU can determine whether there is a ChirpIoT™ pilot signal in the channel by detecting whether the CAD-IRQ signal is pulled high or not in a certain period of time. If it exists, the external MCU can continue to wait for the IRQ interrupt of data reception and then proceed to data reception; if there is no ChirpIoT™ pilot signal, the external MCU can detect if the CAD-IRQ signal is pulled high within a certain period of time.

Transceivers

Then quickly exit the RX mode and wait for the Enter Other Mode command.

In addition, in order to minimize the false triggering of CAD-IRQ, the filtering function of CAD-IRQ has been added, which can be activated by configuring the registers.

address	processor register	BIT	recommended value	fill out or in (information on a form)	clarification
[1][0x25]	REG_DEC_VAL_THRE	1:0	0x3	W/R	CAD detection threshold, which specifies the number of consecutively received preamble after CAD pull-up.

10.3.2 ChirpIoT signaling data segment detection

In order to avoid missing the leading signals of ChirpIoT signals and thus failing to realize effective CAD detection during the actual reception process, the chip also supports the detection of non-ChirpIoT leading signals, i.e., the ChirpIoT data segment signal detection function. By configuring registers REG_SIGNAL_POWER_THRED, REG_DEC_PLD_THRE, REG_DEC_PLD_EN, the data segment signal detection can be realized.

Register Configuration Description:

address	processor register	BIT	recommended value	fill out or in (information on a form)	clarification
[1][0x35]	REG_SIGNAL_POWER_THRED	7:4	4'b1110	W/R	CAD Threshold of valid signals, only energy greater than this value Only then can you enter the judgment of CAD valid signals.
	REG_DEC_PLD_THRE	3:2	2'b10	W/R	CAD Threshold for the number of valid signals, which can only be pulled up if a preamble is received for a number of this value that satisfies the condition CAD.
	REG_DEC_PLD_EN	1	1'b0	W/R	CAD preamble detection enable 0: disable 1: enable

10.4 MAPM Mode Function

In the ChirpIoT star networking scenario, in order to ensure that the node devices can receive the data sent by the gateway, the gateway will use a very long preamble to send, for example, the preamble occupies 1 second. In this case, all node devices will receive preamble until the node network address is displayed in the payload phase. At this time, the software of the node devices can compare the received network address and the node device's own address, if different, it will stop receiving to reduce power consumption, and if it meets the conditions of address reception, it will continue to receive. This program has several problems:

- ① Node devices that do not meet the conditions for address reception will not stop receiving until the payload stage, wasting a lot of power consumption.
- ② Node devices that meet the address receiving conditions also need to receive a full preamble of extra-long time before entering the payload receiving phase, which is also a waste of power consumption.

To solve the above two power consumption problems, the chip adds a new frame structure mode **Multiple Address Preamble Mode**.

Transceivers

(MAPM). The frame structure pattern is to insert a special address in the Preamble phase so that the node end is in the Preamble

The problem ① has been solved by determining in advance whether the address reception condition is satisfied according to the address at the stage. In addition, in order to

To solve problem ②, a variable counter is added to the special address, indicating how many Preamble lengths are still available, and the node-side device that meets the conditions for address reception can enter the low-power sleep state in advance, and then judge how long it needs to enter the reception state in advance according to the counter to ensure that the data is normally received, so as to further save the power consumption of the node-side device that meets the conditions for address reception.

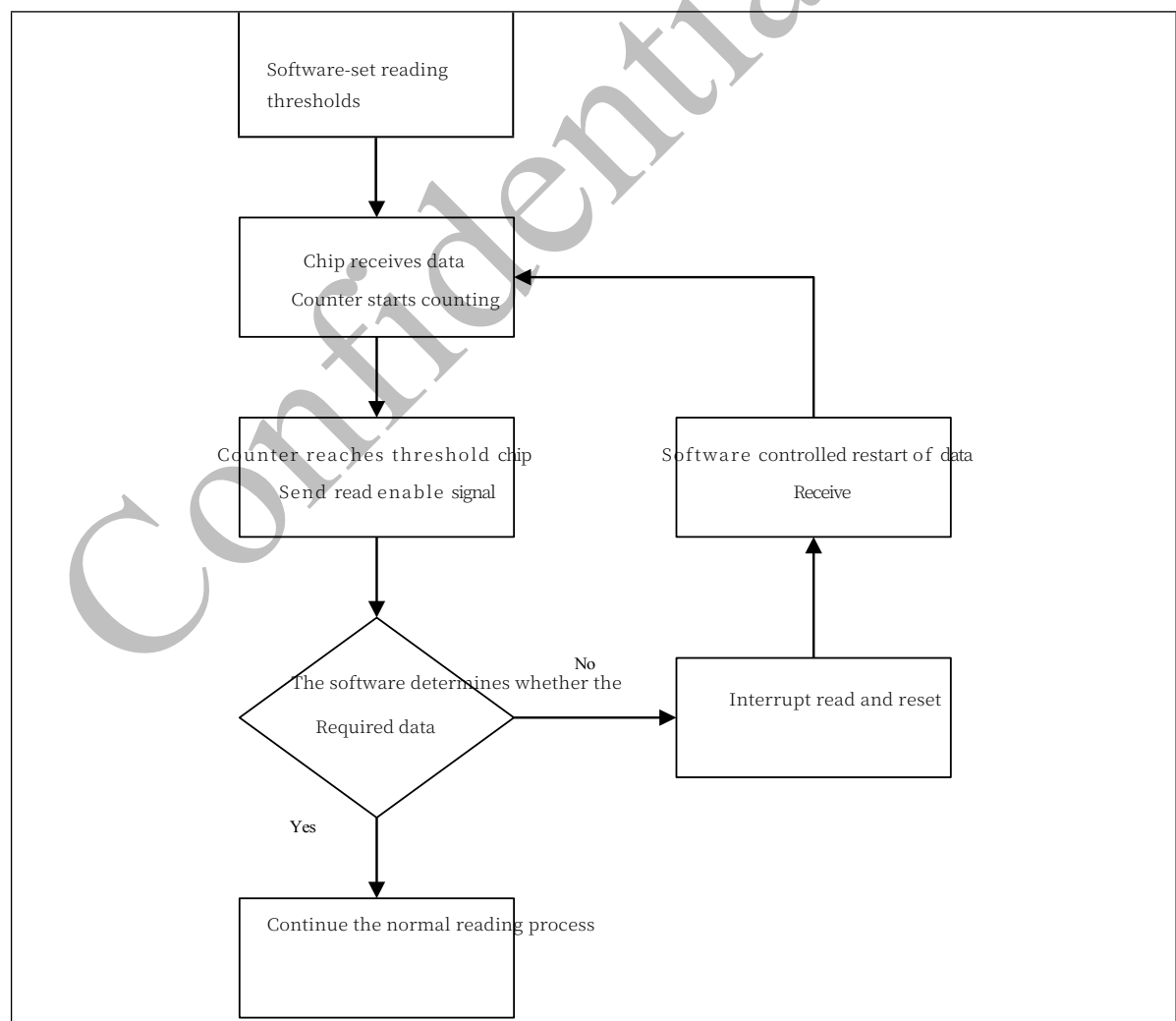
10.5 eFuse Features

The chip supports 82Bytes of eFuse space area for customer use, which can be used to store data without loss of power.

10.6 early discontinuance function

The chip early interrupt function is in the process of the chip reading a frame of data, view the data that has been solved, determine whether it is what you want, and then decide to continue reading or abandon the frame of data.

The flowchart is shown below:



10.7 Intelligent SF Recognition

In order to realize the lightweight gateway device, the chip is designed with an intelligent SF recognition hardware circuit, which can complete the intelligent recognition in the same frequency point according to the SF mode of ChirpLoT™ in the actual channel, and automatically configure the SF parameters of the chip to achieve the purpose of receiving different SF signal data.

The specific implementation flow is shown Figure 10-2:

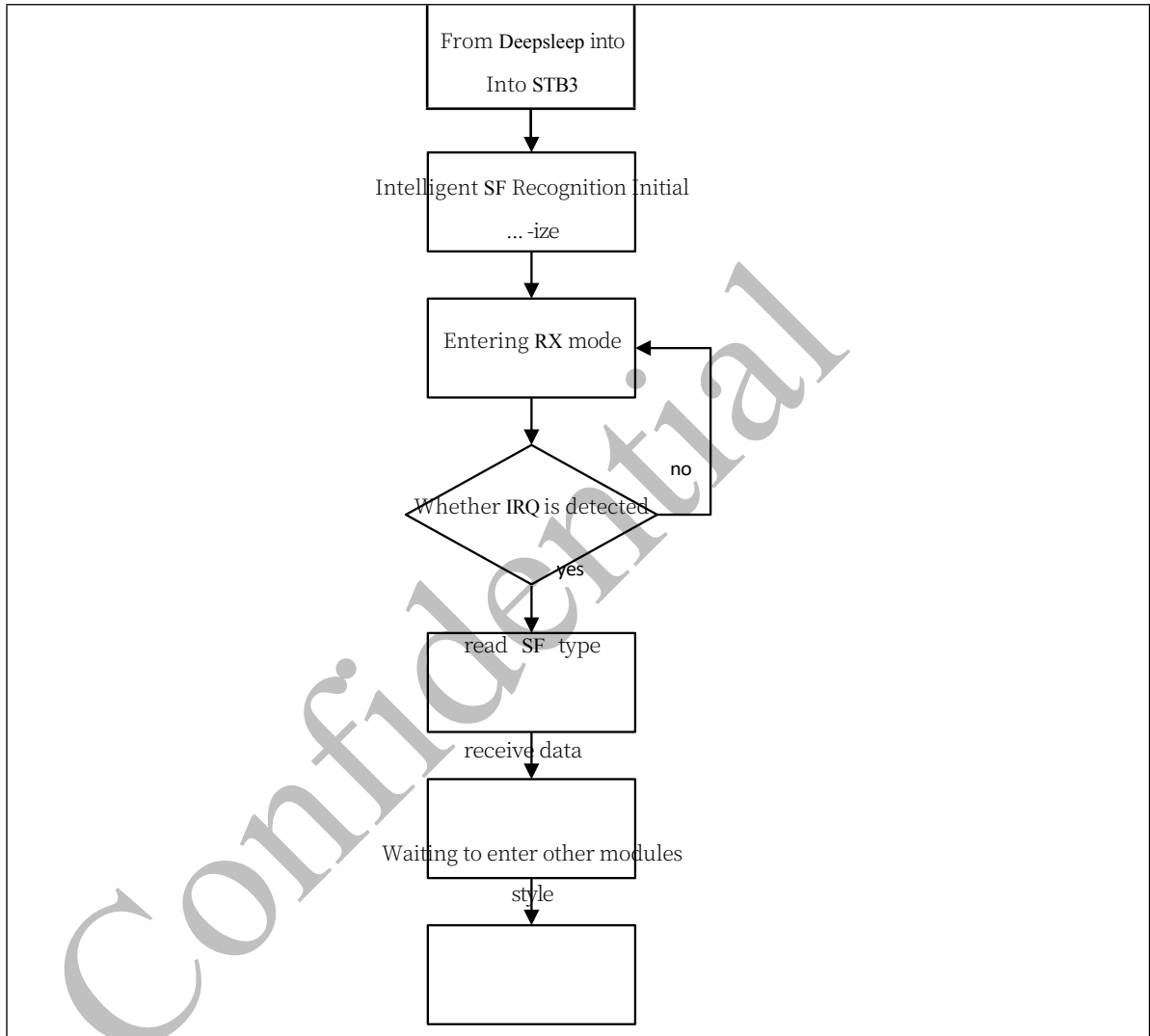
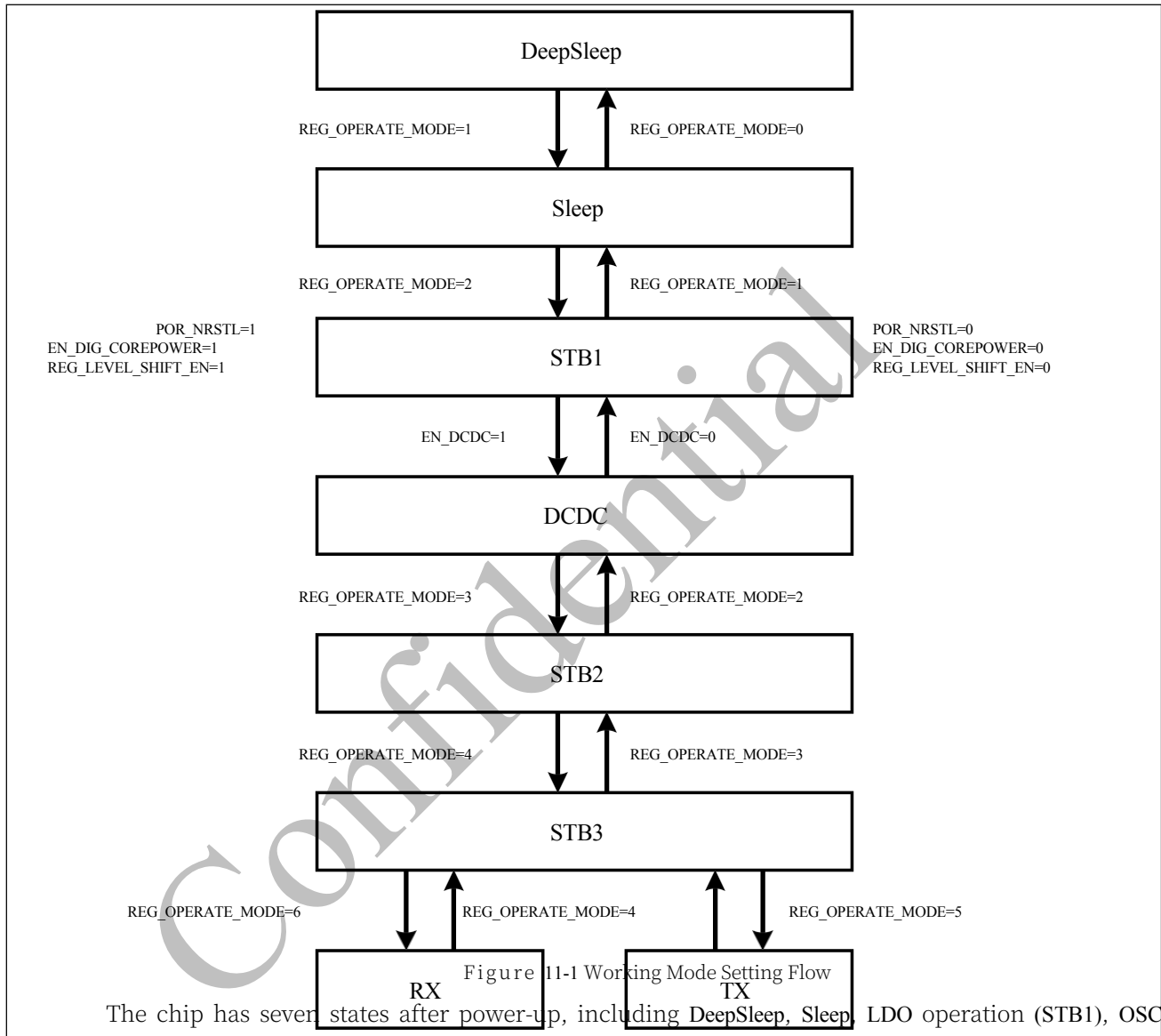


Figure 10-2 Intelligent SF Identification

11 Description of operating modes

11.1 summarize

The chip has total of 7 modes: DeepSleep mode, Sleep mode, STB1 mode, STB2 mode, STB3 mode, TX mode and RX mode.



The chip has seven states after power-up, including DeepSleep, Sleep, LDO operation (STB1), OSC operation (STB2), OSC output (STB3), TX mode, and RX mode. All states can be switched through the configuration registers.

Note: If the peripheral circuitry of the chip does not support DCDC, you can go directly from STB1 to STB2 without configuring EN_DCDC. Refer to Chapter 13 for the chip circuitry.

11.2 Description of operating modes

11.2.1 Deep Sleep Mode

This mode operates in the 3V logic area and the remaining modules are inoperable except for the 3V logic area registers, which are configurable, keeping the chip power consumption to a minimum.

11.2.2 Sleep mode

This mode enables low-power LDOs, low power consumption of the chip, and can support power preservation of all registers, but prohibits configuration of registers other than the 3V logic region via SPI/I2C.

11.2.3 STB1 Mode

This mode is when the high-performance LDO starts operating and the low-voltage region logic is enabled.

11.2.4 DCDC model

The chip's DCDC function can be turned on by writing high to the EN_DCDC register in the STB1 state.

Note: DCDC mode requires related peripheral circuit support, please refer to Chapter 13.

11.2.5 STB2 mode

This mode turns on the OSC crystal, the OSC starts to work to maintain the oscillation but does not output to other modules, the power consumption is relatively small.

11.2.6 STB3 Mode

In this mode the OSC clock is output to the modules and the chip starts to work normally.

11.2.7 TX Mode

This mode has turned on all TX Transmit related modules for encoding and transmitting packet data.

11.2.8 RX Mode

This mode has turned on all RX receive related modules waiting for packet data to be received and decoded.

12 processor register

The registers of the chip are divided into 4 pages, each page has 128 register addresses, and the register page selection is made by REG_PAGE_SEL of register 0x00, in which the registers with addresses 0x00~0x05 can be operated in any one of the register pages. The following table describes the location and function of each register on the chip.

Attention:

- Read and write operations to the registers in the *Reserved* section are prohibited.
- All registers in the low voltage area support operation in *STB1*, *STB2*, *STB3*, *Tx* and *Rx* modes only.

12.1 System Configuration

page	addresses	processor register	BIT	recommended value	fill out or in (information on a form)	clarification
0/1/2/3	0x00	REG_SOFT_RST	7	0x00	W/R	Soft reset, all except 1.2V configuration registers 1.2V Logic Reset 0: Reset 1: No reset
		Reserved	6:2		W/R	Reserved
		REG_PAGE_SEL	1:0		W/R	0: Select Page0 1: Select Page1 2: Select Page2 3: Select Page3
	0x01	FIFO READ/ WRITE ACCESS POINT	7:0	-	W/R	FIFO Read/Write Address

12.2 Mode Configuration

page	addresses	processor register	BIT	recommended value	fill out or in (information on a form)	clarification
0/1/2/3	0x02	Reserved	7:3	0x00	W/R	Reserved
		REG_OPERATE_MODE	2:0		W/R	0: Deep sleep mode 1: Sleep mode 2: STB1 mode (LDO operation) 3: STB2 mode (OSC operation) 4: STB3 mode (OSC output) 5: Tx Mode 6: Rx Mode
		Reserved	7:6		W/R	Reserved
		EN_LS_3V	5		W/R	1: Level shift enable



PAN3029 / PAN3060 Series Low Power Long Range Wireless

Transceivers					
	0x04	POR_RSTL	4		W/R
		Reserved	3:0	W/R	Reserved

Transceivers
12.3 MAC Configuration (Low Voltage Area Register)

page	addresses	processor register	BIT	recommended value	fill out or in (information on a form)	clarification
0	0x54	Reserved	7:3	0x00	W/R	Reserved
		REG_TX_FIFO_FULL	2		W/R	0: TX FIFO not full 1: TX FIFO full
		REG_TX_FIFO_FULL_SEL	1		W/R	0: select fifo wr done 1: select reg tx fifo full
		Reserved	0		W/R	Reserved
	0x58	REG_RX_TIMEOUT_IMMED	7	0x0F	W/R	1: Rx timeout IRQ generated immediately
		REG_MAPM_IRQ_MASK	6		W/R	IRQ MASK enable in MAPM mode 0: No IRQ interrupt for MAPM is generated 1: Generate IRQ interrupt for MAPM
		Reserved	5		W/R	Reserved
		REG_RX_PLHD_DONE_MASK	4		W/R	Early Interrupt Mode Mask 0: No early interrupt mode IRQ interrupts are generated 1: Generate IRQ interrupts for early interrupt mode
		REG_RX_DONE_MASK	3		W/R	rx complete irq mask 0: Do not generate rx-complete IRQ interrupt 1: Generate IRQ interrupt for rx completion
		REG_RX_PL_CRC_ERR_MASK	2		W/R	Rx payload crc error irq mask 0: IRQ that does not generate Rx payload crc error disruptions 1: IRQ generating Rx payload crc error disruptions
		REG_RX_TIMEOUT_MASK	1		W/R	Rx timeout irq mask 0: IRQ interrupt without Rx timeout 1: IRQ interrupt that generates Rx timeout
		REG_TX_DONE_MASK	0		W/R	Tx completion irq mask 0: IRQ interrupt for Tx completion is not generated 1: Generate IRQ interrupt for Tx completion
	0x5b	Reserved	7:3	0x03	W/R	Reserved
		REG_RX_DONE_RST_TEST	2		W/R	1: Once RX is complete, the modem resets immediately. 0: IRQ is cleared and RX is complete before the modem resumes. classifier for honorific people
		REG_RX_CR_ERR_RST	1		W/R	1: rx code rate error reset enable
		REG_RX_HEADER_ER_RST	0		W/R	1: rx packet header error reset enable

Transceivers					
0x5d	Reserved	7:1	0x00	W/R	Reserved
	REG_2POINT_CAL_EN	0		W/R	Two-point cal enable, active high
	EFUSE_WR_IRQ	7	0x00	R	eFuse Read/Write Finish Interrupt 1: High level indicates read/write completion
	REG_MDM_MAPM_IRQ	6		R	MAPM Interrupt Status Register 1: MAPM IRQ; 0: Non-MAPM IRQ
	Reserved	5		W/R	Reserved
	REG_RX_PLHD_DONE_IRQ	4		W/R	Write 1 Clear early interrupt
	REG_RX_DONE_IRQ	3		W/R	Write 1 clear rx done irq
	REG_RX_PL_CRC_ERR_IRQ	2		W/R	Write 1 clear rx payload crc error irq
0x6c	REG_RX_TIMEOUT_IRQ	1		W/R	Write 1 clear rx timeout irq
	REG_TX_DONE_IRQ	0		W/R	Write 1 clear tx done irq
	0x6D REG_MAPM_PN_CNT	7:0	0x00	R	MAPM mode preamble quantity counter
	0x6E REG_MAPM_ADDRX	7:0	0x00	R	MAPM mode addrx

12.4 GPIO Interface Configuration (Low Voltage Register)

page	address	processor register	BIT	recommended value	fill out or in (information on a form)	show
0	0x5e	REG_OUT_FUNC_SEL_PA	7	0x00	W/R	1: Use as normal IO 0: Used as external PA
		REG_OUT_FUNC_SEL_CAD	6		W/R	1: Use as normal IO 0: Used as CAD pin
		Reserved	5:0		W/R	Reserved
	0x5f	GPIO_PDEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 pull-down enable GPIO_PDEN LSB[0]: GPIO0 pull down enable Other Configuration Reserved
	0x60	Reserved	7:4	0x00	W/R	Reserved
		GPIO_PDEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 pull-down enable GPIO_PDEN MSB[2]: GPIO10 pull-down enable Other Configuration Reserved
	0x61	GPIO_PUEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 Pull-up Enable GPIO_PDEN LSB[0]: GPIO0 pull-up enable Other Configuration Reserved
	0x62	Reserved	7:4	0x00	W/R	Reserved
		GPIO_PUEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 Pull-up Enable GPIO_PDEN MSB[2]: GPIO10 Pull-up Enable Other Configuration Reserved
	0x63	GPIO_DIEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 input enable GPIO_PDEN LSB[0]: GPIO0 input enable Other Configuration Reserved
		Reserved	7:4		W/R	Reserved

Transceivers						
0x64	GPIO_DIEN MSB	3:0	0x00	W/R	GPIO_PDEN MSB[3]: GPIO11 input enable GPIO_PDEN MSB[2]: GPIO10 input enable Other Configuration Reserved	
0x65	GPIO_OE LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 output enable GPIO_PDEN LSB[0]: GPIO0 output enable Other Configuration Reserved	
0x66	Reserved	7:4	0x00	W/R	Reserved	
	GPIO_OE MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 output enable GPIO_PDEN MSB[2]: GPIO10 Output Enable Other Configuration Reserved	
0x67	GPIO_OUT LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 outputs GPIO_PDEN LSB[0]: GPIO0 outputs Other Configuration Reserved	
0x68	Reserved	7:4	0x00	W/R	Reserved	
	GPIO_OUT MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 outputs	
					GPIO_PDEN MSB[2]: GPIO10 outputs Other Configuration Reserved	
0x74	Reserved	7:6	-	R	Reserved	
	GPIO_IN LSB	5:0	-	R	GPIO_PDEN LSB[3]: GPIO3 Inputs GPIO_PDEN LSB[0]: GPIO0 input Other Configuration Reserved	
0x75	Reserved	7:6	-	R	Reserved	
	GPIO_IN MSB	5:0	-	R	GPIO_PDEN LSB[3]: GPIO11 inputs GPIO_PDEN LSB[2]: GPIO10 Inputs Other Configuration Reserved	

12.5 Basic Operation Configuration

page	addresses	processor register	BIT	recommended value	fill out or in (information on a form)	clarification
	0x06	Reserved	7:3	0x00	W/R	Reserved
		REG_TX_CFG_MODE	2		W/R	0: Single transmission mode 1: Continuous transmission mode
		REG_RX_CFG_MODE	1:0		W/R	0: Single reception mode 1: Single reception mode with timeout 2: Continuous reception mode
	0x07	REG_RX_TIMEOUT[7:0]	7:0	0x00	W/R	Receive timeout setting, lower 8 bits in microseconds
	0x08	REG_RX_TIMEOUT[15:8]	7:0	0x00	W/R	Receive timeout setting, high 8 bits, in microseconds
		BW	7:4		W/R	Bandwidth: 6: 62.5 kHz 7: 125 kHz 8: 250 kHz

Transceivers					
3	0x0D			0x98	9: 500 kHz
		CODING RATE	3:1	W/R	Code rate: 1: 4/5 2: 4/6 3: 4/7 4: 4/8
		Reserved	0	W/R	Reserved
	0x0E	SPREADING FACTOR	7:4	0x78	W/R SF: 5: 32 chips / symbol 6: 64 chips / symbol 7: 128 chips / symbol 8: 256 chips / symbol 9: 512 chips / symbol 10: 1024 chips / symbol 11: 2048 chips / symbol 12: 4096 chips / symbol
		RXPAYLOADCRC CON	3		W/R 0: CRC off 1: CRC open
		Reserved	2:0		W/R Reserved
	0x0F	SYNCWORD	7:0	0x12	W/R synchronized characters
	0x10	IF[7:0]	7:0	0xC0	W/R IF Configuration, Sub-High 8-bit
	0x11	IF[11:8]	7:4	0x0F	W/R IF Configuration, High 4-Bit
		Reserved	3:0		W/R Reserved
	0x12	Reserved	7:4	0x06	W/R Reserved
		LOWDATARATE	3		W/R 0: Normal mode 1: Low rate mode
		Reserved	2:0		W/R Reserved
	0x13	PREAMBLE_LEN[7:0]	7:0	0x08	W/R preamble length 8 bits lower
	0x14	PREAMBLE_LEN[15:8]	7:0	0x00	W/R preamble length 8 bits high
	0x1A	SPI3_EN_REG	7	0x03	W/R 3-Wire SPI Select Register 0: 3-wire SPI mode is not selected; 1: Select 3-wire SPI mode
		REG_MOSI_PUEN	6		W/R Register control mode for mosi pins puen control registers depositor (computing)
		REG_CSN_PUEN	5		W/R Register control mode for csn pins csn control registers tool
		REG_SCK_PUEN	4		W/R Register control mode for sck pins sck control registers tool
		REG_IN_PAD_MODE	3		W/R SPI input pin manual configuration mode enable
		REG_OUT_PAD_MODE	2:0		W/R SPI's MISO pin mode selection for 1-to-Many SPI Applications
	0x24	Reserved	7:4	0x00	W/R Reserved
		EN_DCDC_3V	3		W/R DCDC Enable 0: DCDC bypass 1: DCDC work

Transceivers

	0x26	Reserved	2:0	0x00	W/R	Reserved
		XTAL_ACTIVE_EN_3V	7		W/R	Supports 3V powered active crystal channel enable to ensure that the chip can operate at the crystal 3V clock 1: open; 0: close
		XTAL_STARTUP_EN_3V	6		W/R	Crystal fast start enable. 1: on; 0: off.
		EN_DIG_COREPOWER_3V	5		W/R	Digital core voltage enable switch 1: open; 0: close
		Reserved	4:0		W/R	Reserved

Confidential

13 reference schematic

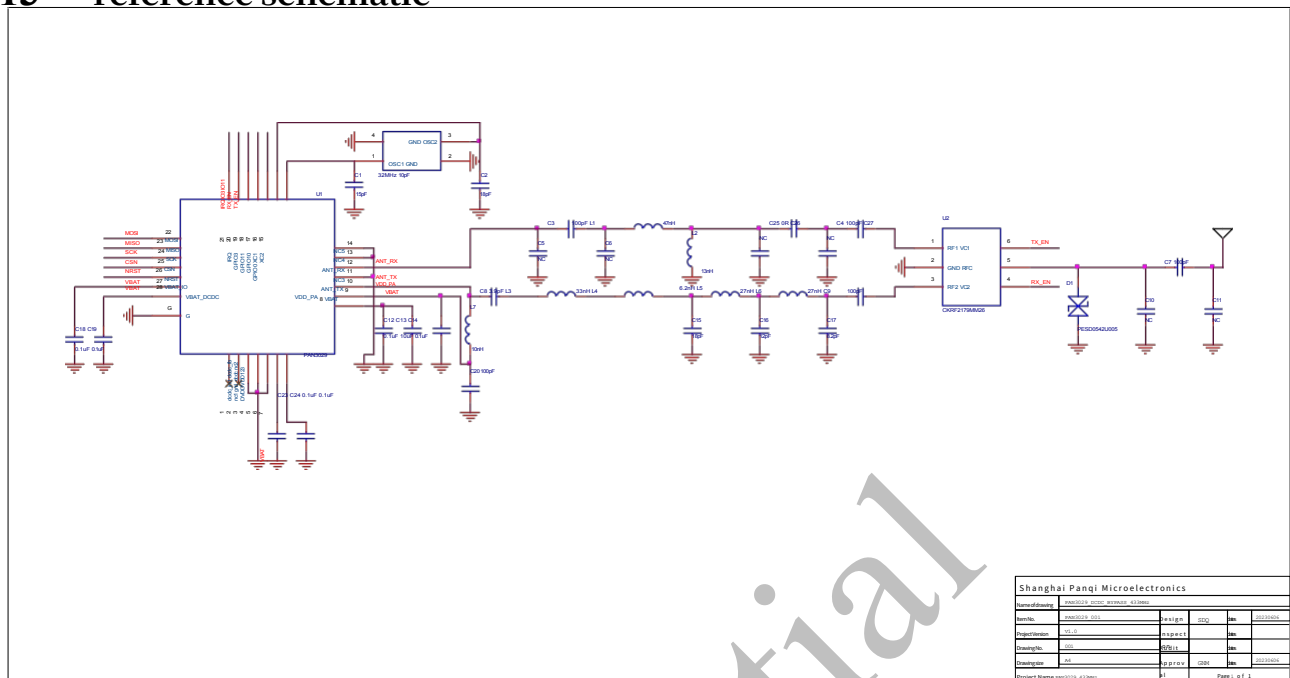


Figure 13-1 LDO Mode 433MHz Reference Schematic

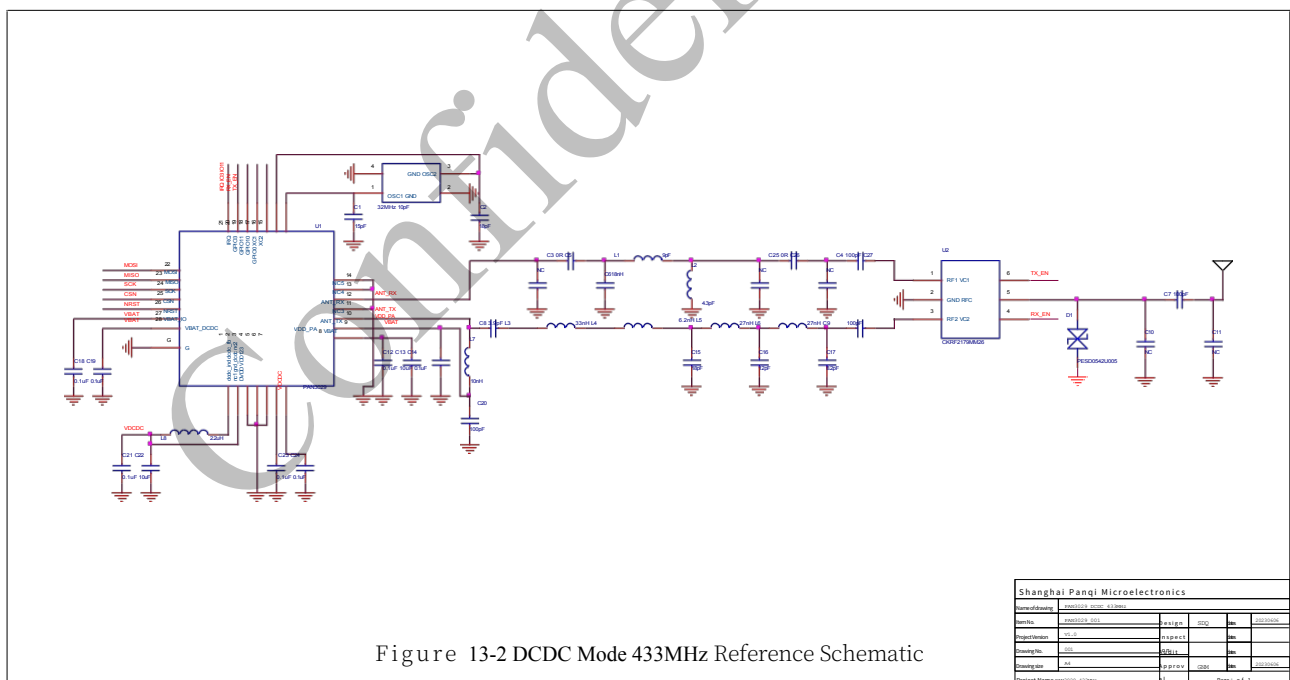
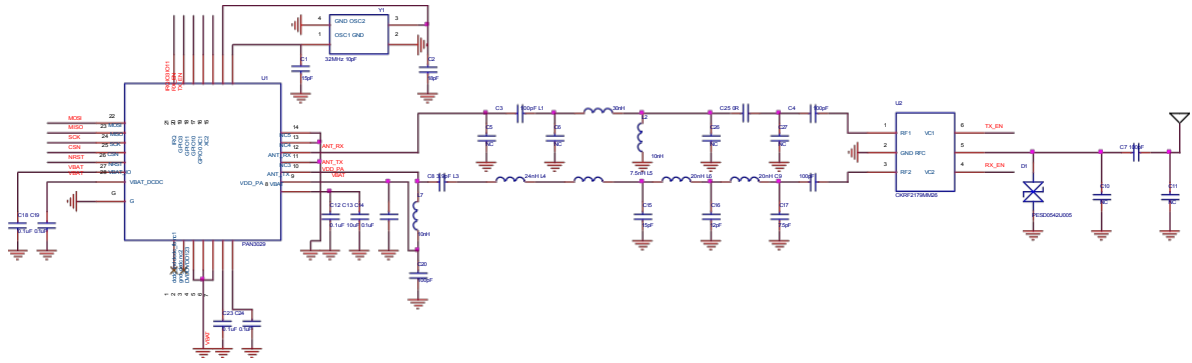
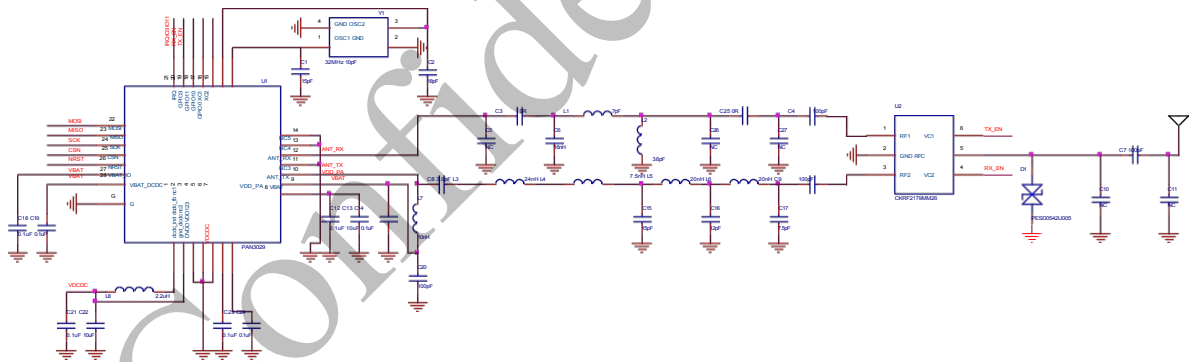


Figure 13-2 DCDC Mode 433MHz Reference Schematic



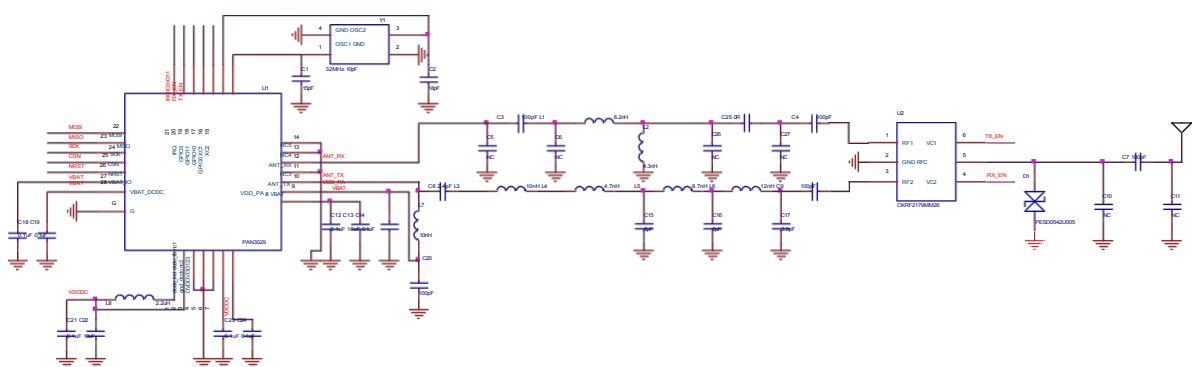
Shanghai Panqi Microelectronics					
Name of Drawing: PAN3029_2020_010001_010001					
Version	1.0.0.0	Project Name	PAN3029	Project No.	PAN3029
Author	Wang Jia	Project Manager	Wang Jia	Project No.	PAN3029
Reviewer	Wang Jia	Project Manager	Wang Jia	Project No.	PAN3029
Designer	Wang Jia	Project Manager	Wang Jia	Project No.	PAN3029
Project Name: PAN3029_2020_010001_010001					
Page: 1 of 1					

Figure 13-3 LDO Mode 490MHz Reference Schematic



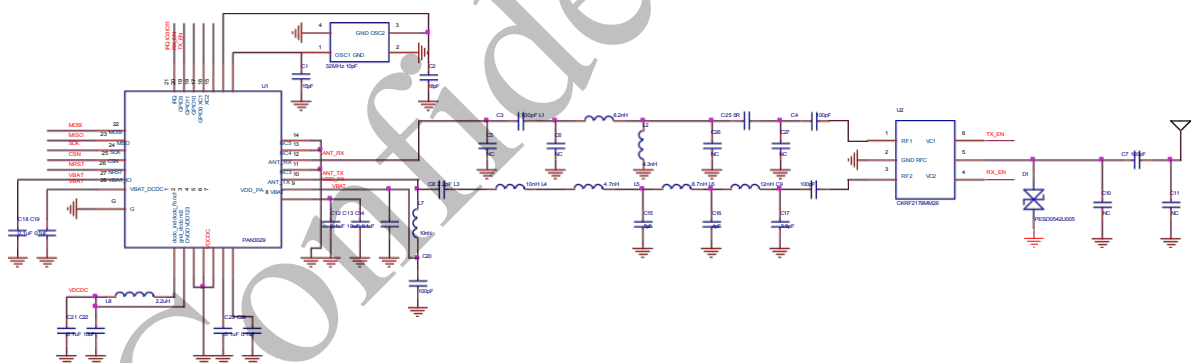
Shanghai Panqi Microelectronics					
Name of Drawing: PAN3029_2020_010001_010001					
Version	1.0.0.0	Project Name	PAN3029	Project No.	PAN3029
Author	Wang Jia	Project Manager	Wang Jia	Project No.	PAN3029
Reviewer	Wang Jia	Project Manager	Wang Jia	Project No.	PAN3029
Designer	Wang Jia	Project Manager	Wang Jia	Project No.	PAN3029
Project Name: PAN3029_2020_010001_010001					
Page: 1 of 1					

Figure 13-4 DCDC Mode 490MHz Reference Schematic



Shanghai Panqi Microelectronics						
Name of drawing		PMS3023-96 (2000)				
Item No.	YMK3023-001	Design	SDQ	Rev	J2030326	
Project Version	V5.0	Inspect		Rev		
Drawing No.	001	Drawn by		Rev		
Drawing size	A4	Approved	Q2H	Rev	J2030326	
Project Name		PMS3023-96 (2000)		Page 1 of 1		

Figure 13-5 863MHz Reference Schematic



Shanghai Panqi Microelectronics						
Name of drawing		PANQI2020-010000				
Item No.	PANQI2020-0101	Design	CCQ	Rev	2020-03-04	
Project Version	V5.0	Aspect		Rev		
Drawing No.	001	20/01/11		Rev		
Drawing Date	01/01/11	Approval	CCQ	Rev	2020-03-04	
Product Name: PANQI2020-010000		Alt	Page 1 of 1			

Figure 13-6 902MHz Reference Schematic

14 Package Size

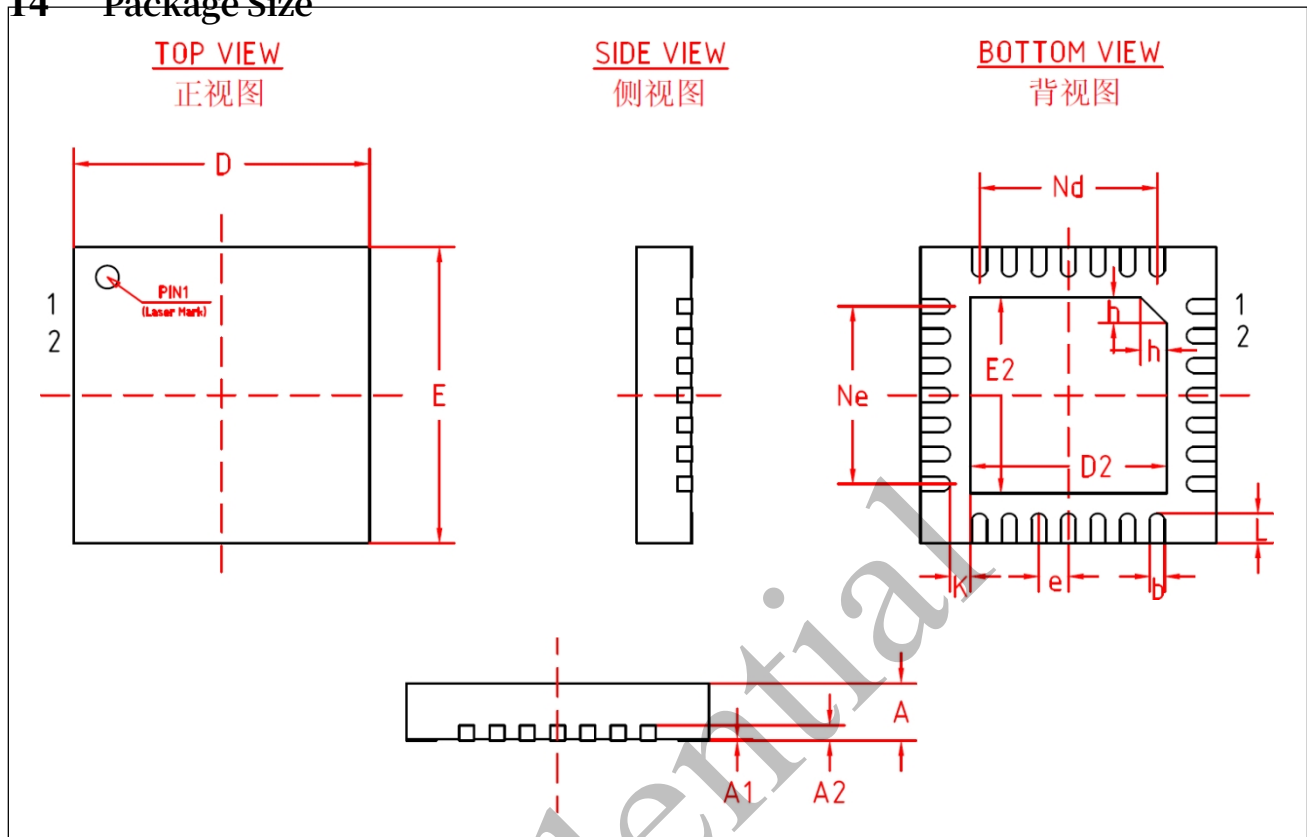


Figure 14-1 QFN28 Package
Diagram Table 14-1 QFN28
Package Dimensions

notation	Minimum value (mm)	Typical value (mm)	Maximum value (mm)
A	0.70	0.75	0.80
A1	-	0.02	0.05
A2	0.203 REF		
b	0.15	0.20	0.25
D	3.90	4.00	4.10
D2	2.55	2.65	2.75
E	3.90	4.00	4.10
E2	2.55	2.65	2.75
e	0.40 BSC		
K	0.225	0.275	0.325
L	0.35	0.40	0.45
h	0.30	0.35	0.40
Ne	2.40 BSC		
Nd	2.40 BSC		

15 caveat

- 1) This product is a CMOS device and should be protected from static electricity during storage, transportation, and use.
- 2) The device should be well grounded when used.
- 3) Reflow temperature should not exceed 260°C. The lead-free reflow process curve is shown below:

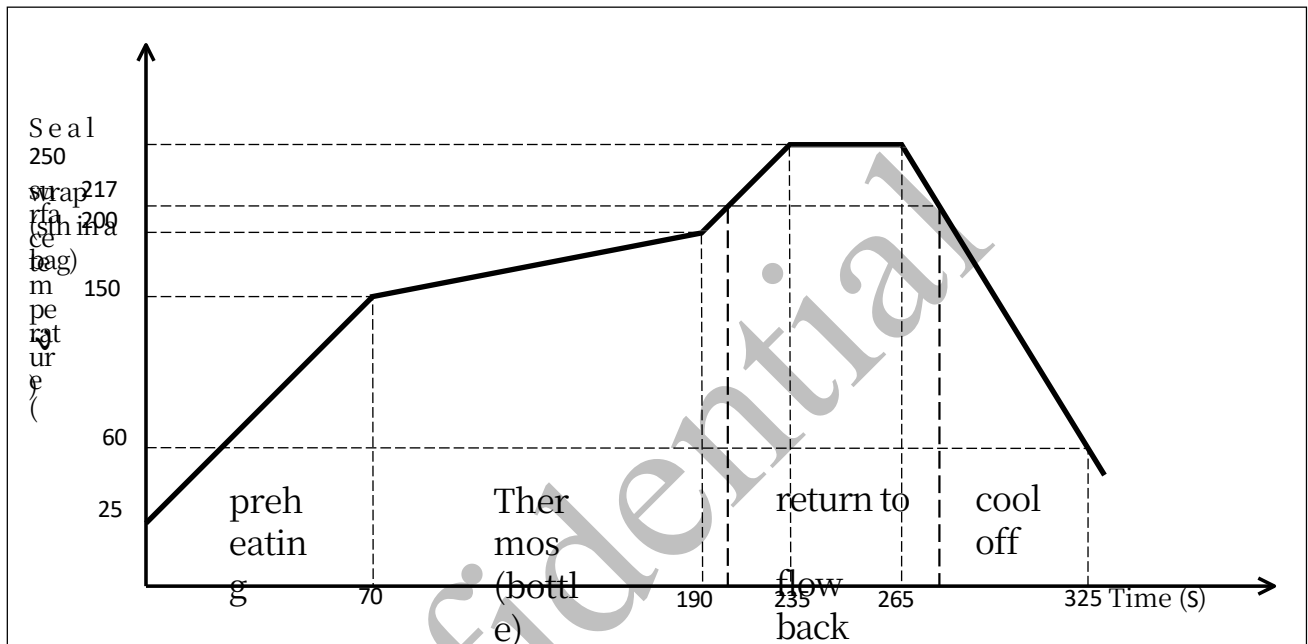


Figure 15-1 Reflow Soldering Process Profile

16 Storage conditions

- 1) Storage in sealed packages: up to 12 at temperatures less than 30°C and humidity less than 90%.
- 2) After the bag is opened, the components must be conformed when they will be used in the reflow process or other high temperature processes:
 - a) Completion within 72 hours and in a factory environment of less than 30°C ≤ 60%RH;
 - b) Store at 10%RH;
 - c) Bake at 125°C for 24h to remove internal moisture before use.
- 3) MSL (package moisture sensitivity): Level 3 (as determined by IPC/JEDEC J-STD-020)

Confidential

Transceivers abbreviations

ADC	analog-to-digital converter
CAD	channel activity detection
Chirp	linear frequency modulation (LFM)
CRC	cyclic redundancy check
CSN	SPI Chip Select Signal
DAC	digital-to-analog converter
DCDC	DC converter
FIFO	first in, first out (FIFO)
GPIO	General purpose inputs and outputs
IRQ	interrupt request
LDO	Low Dropout Linear Regulators
LPF	low-pass filter
MAC	MAC layer
MCU	microprocessor
Mixer	mixer
Modem	modem
OSC	oscillators
PA	power amplifier
RF	a radio frequency
PLL	phase-locked ring
PMU	Power Management Unit
POR	Power-on reset
RAM	random access memory (RAM)
RSSI	signal strength indication
SCK	SPI Clock Signal
SF	spreading factor
SPI	Serial Peripheral Interface
STB	standby mode



Sync	Transceivers	synchronization
VCO		voltage controlled
		oscillator

revision history (of a document, web page etc)

Version	Date	Content
V1.0	2023.05	first edition
V1.1	2023.06	Add packaging method, reference schematic, optimize pin1 and pin2 name in pin definition, no change in function.
V1.2	2023.07	Addition of PAN3060 Series, Electrical Characteristics, etc.
V1.3	2023.08	Updated reference schematic with additional operating voltage description
V1.4	2023.09	Supplementary reference schematics for different frequency bands and supplementary electrical characteristics for different frequency bands

documentation

The contents of this document may be updated from time to time due to version upgrades or other reasons. Unless otherwise agreed, the contents of this document are intended as a guide to use only, and all statements, information and recommendations in this document do not constitute any warranty, express or implied.

logo

Panqi is a trademark of Panqi Microelectronics Ltd. Other names mentioned in this document are trademarks/registered trademarks of their respective owners.

statement denying or limiting responsibility

All or some of the products, services or features described in this document may be excluded from your purchase or use. Unless otherwise contractually agreed, Granite Microelectronics Ltd. makes no representations or warranties, express or implied, with respect to the contents of this document.

联系方式



上海磐启微电子有限公司

张江高科技园区盛夏路 666 号 D 栋 302 室
上海市浦东新区



021-50802371

<http://www.panchip.com>



Confidential